2 Lab 1: Tyres

2.1 Introduction

The primary objective of this laboratory is to investigate the complex relationship between tyre behavior and vehicle performance. Tyres represent the only contact point between the vehicle and the ground; therefore, understanding their thermal and frictional characteristics is fundamental for optimizing vehicle dynamics.

2.2 Exercise 1: Analysis of Tyre Thermal Behavior

The first part of this laboratory focuses on the study of a **tyre thermal model** to understand how temperature variations affect a vehicle's grip and overall performance. Through a simulation performed in a MATLAB environment, we analyze a full lap on a flat oval circuit, specifically the **Indianapolis Motor Speedway**.

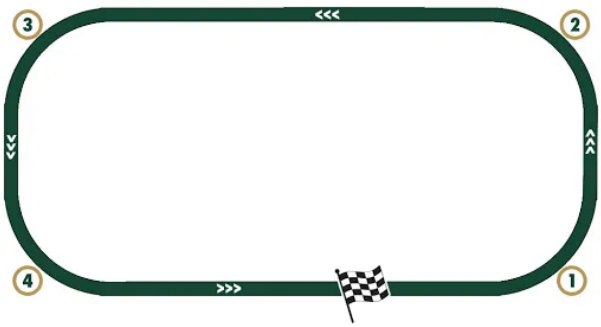


Figure 13: Track Layout

The main goal is to evaluate how specific setup parameters and environmental factors influence the heat generation and dissipation of the tires.

2.2.1 Code Logic

- `mu.mat` **file**
  The file contains the following variable vectors:

  1. `T`: tire temperature range (1x301 double)
  2. `v`: tire sliding velocity range (1x108 double)
  3. `mumat`: Effective friction coefficient matrix (108x301 double).
     Each row corresponds to a specific sliding velocity `v` while each column corresponds to a specific temperature `T`.

  In reality, the tire's grip level is not fixed but depends on the tire's temperature and how much it's sliding on the asphalt. The sliding speed directly sets the excitation frequency of the rubber as it interacts with the road macro-roughness

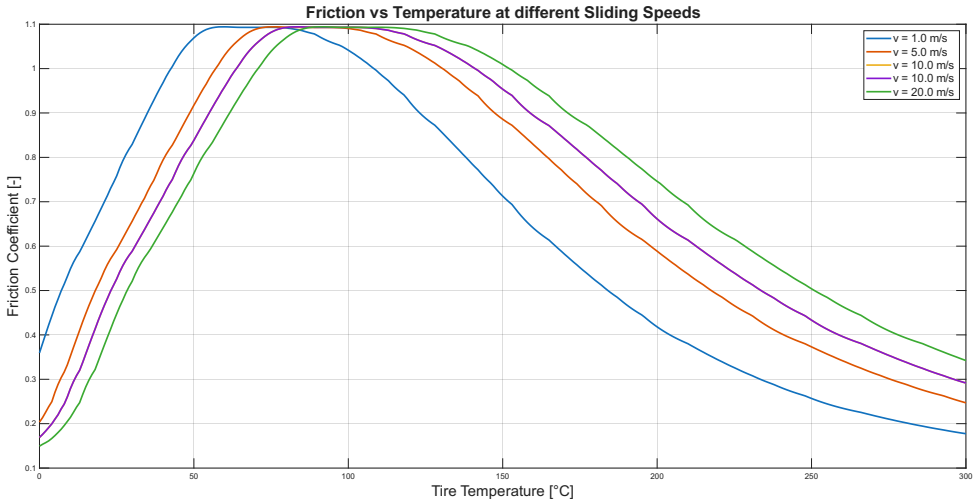


Figure 14: WLF Master Curve at different Sliding Speeds

The `mu.mat` file represents s an experimental map (`Lookup Table`) which tells the simulator the grip available at a certain tire temperature and sliding speed.

The model simplifies the 3D friction map (μ vs. speed and temperature) by extracting a single row from the `mumat` matrix corresponding to a reference sliding speed of v = 5 m/s. Consequently, the simulation assumes the available grip depends exclusively on the dynamic variation of tire temperature.

```
Vref                = 5;              % [m/s] Reference sliding speed
i1=find(v>Vref,1,'first');            % Use friction at reference
    slideing speed
Track.Mu.T          = T+273;          % [K] Tempertaure vector
Track.Mu.mu         = mumat(i1,:);    % [] Friction vector
```

- `Indianapolis.mat` **file**
  The file contains the layout of the Indianapolis circuit.
  The first column `Indianapolis(:,1)` contains the different sectors lengths of the circuit. The second column `Indianapolis(:,2)` represents the curvature radius R.
  For simulation purposes, the track curvature ($\gamma = 1/R$) is derived:

```
Track.Sectors.L = Indianapolis(:,1)';
Track.Sectors.Gamma = 1./Indianapolis(:,2)';
Track.Sectors.Gamma(find(Track.Sectors.Gamma == Inf)) = 0;
```

Rather than using exceedingly large numbers or 'Infinity' to represent straight sections curvature radii, track dataset assign a "dummy" value of $R = 0$. Consequently, computing the inverse ($1/R$) yields non-finite values (Inf in MATLAB) for these sectors. The algorithm actively identifies these Inf values and overwrites them with $\gamma = 0$, correctly defining the straight sections.

To perform the dynamic vehicle simulation, the track dataset provided must be converted into a continuous, high-resolution spatial profile. The algorithm reconstructs the track by building a step-like function of the track properties against the curvilinear abscissa (s).

```
Track.ds=1;
% Sector 1 distance and curvature definition
Sv(1)=0;
Sv(2)=Track.Sectors.L(1);
Gv(1)=Track.Sectors.Gamma(1);
Gv(2)=Track.Sectors.Gamma(1);
% Loop through the remaining sectors to build the step-function
    profile
for c1=2:length(Track.Sectors.L)
    Sv(2*c1-1)=Sv(2*c1-2)+Track.ds;
    Sv(2*c1)=Sv(2*c1-1)+Track.Sectors.L(c1);
    Gv(2*c1-1)=Track.Sectors.Gamma(c1);
    Gv(2*c1)=Track.Sectors.Gamma(c1);
end

Track.Traj.s=0:0.1:sum(Track.Sectors.L);
Track.Traj.gamma=interp1(Sv,Gv,Track.Traj.s);
Track.L  = max(Track.Traj.s);
```

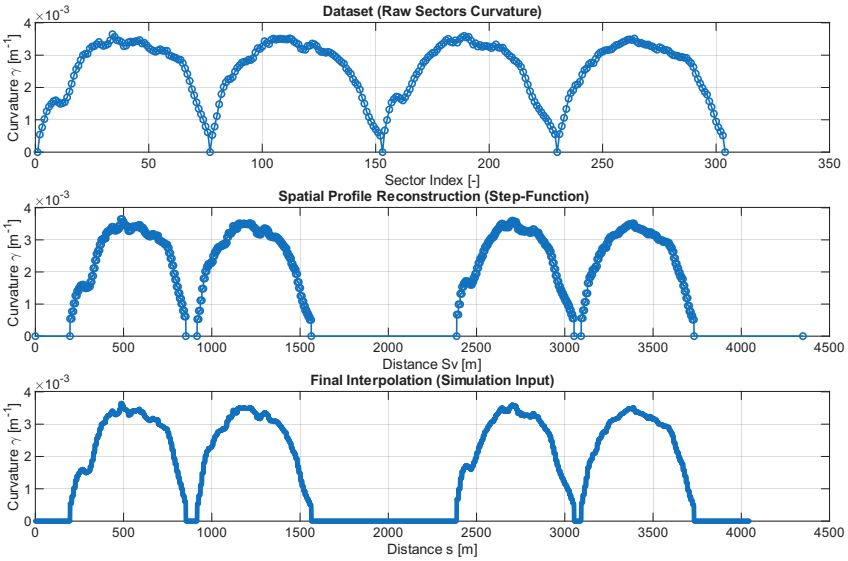


Figure 15: Track Curvature Discretization and Interpolation Process

The resulting continuous arrays provide the numerical solver with smooth, queryable track inputs at any given point along the trajectory, ensuring computational stability during the integration process.

From the perspective of the vehicle's equations of motion, the track is only perceived as a $\gamma(s)$ profile. For each spatial coordinate s, the solver extracts the corresponding curvature value and computes the tire's and vehicle's dynamic response. The solver does not behave like a "path-follower" system, meaning that the 2D Cartesian trajectory (X,Y) is completely strictly not required for simulation purposes.

- **`ode15s` solver**
  The core of the simulation is the MATLAB `ode15s` function, a variable-step solver.

```
[t,x] = ode15s(@(t,x) TMMT(t,x,Vehicle,Track,Tyre,Air,g), 0:.01:
    Tend, [0 T0 T0 T0 T0 Vehicle.Vmax]);
```

  The solver computes the evolution of the 6-dimensional state vector x over time:

  1. x_1: Vehicle position along the track center-line, s [m]
  2. x_2, x_4: Front and rear tire tread temperatures, T_{tread} [°C]
  3. x_3, x_5: Front and rear tire carcass temperatures, $T_{carcass}$ [°C]
  4. x_6: Vehicle longitudinal speed, v [m/s]

  The solver operates in the time domain (t) in synergy with the `TMMT` function, which evaluates the state derivatives ($\dot{x}$) at each integration step. A key engineering feature of this architecture is the **kinematic coupling** used to drive the time-based solver through the space-defined track.

  By defining the derivative of the position as the current velocity ($\dot{x}_1 = v$), `ode15s` updates the curvilinear abscissa s at every step. This allows the `TMMT` function to query the track database (curvature and friction) based on the current s, effectively "driving" the vehicle model along the physical circuit.

- **`TMMT` function**
  The Thermal Mechanical Model of Tyre (TMMT) acts as the core physics engine of the vehicle simulation. Given a specific point in time, it takes the current state variables—namely the vehicle's position, actual speed, and the temperatures of the tire treads and carcasses—and computes their respective time derivatives.

```
function [xp,Qdef,QconvTA,QcondTT,Qfric,mu]=TMMT(t,x,Vehicle,Track,
    Tyre,Air,g)
```

  1. **Vehicle Position along the track**
     The code starts from defining the current vehicle position along the track:

```
%% Determine values for track curvature and friction
xin=x(1);  % position along track
if xin>Track.Traj.s(end) % restart position if more laps are
    completed
    xin=x(1)-Track.Traj.s(end)*floor(x(1)/Track.Traj.s(end));
end
```

     Multiple laps simulations can be handled; if the current position is higher than one single lap length, the code resets the vehicle position making it starting from zero again, simulating the start/finish line passage.

  2. **Friction Coefficient Computation**
     Based on the current position along the track, the code extracts the corresponding curvature value γ. The current friction values for the front and rear axles are extracted by interpolating the $\mu - T$ diagram at the given front and rear T_{tread}.

```
% track curvature
gamma = abs(interp1(Track.Traj.s,Track.Traj.gamma,xin,'spline')
    );
mu(1) = interp1(Track.Mu.T,Track.Mu.mu,x(2)); % front tires mu
mu(2) = interp1(Track.Mu.T,Track.Mu.mu,x(4)); % rear tires mu
mum = mean(mu);
```

A mean friction coefficient value `mum` between the front and rear axle is used to compare it with the grip limit available.

A peculiar mathematical singularity arises whenever the speed of the vehicle has to be derived for high-speed, low-curvature sections (e.g., very wide corners or straights).

The equation for the vehicle limit speed is mathematically expressed as:

$$V_{lim} = \sqrt{\frac{m \cdot g}{\frac{m \cdot \gamma}{\mu_m} + C_{lA}}}$$

In gentle/null curves, the downforce can grow faster than the centrifugal demand which physically means that the vehicle has virtually infinite grip and will never slide outward; its speed is solely limited by engine power.

Numerically, however, this creates a critical issue. If the available friction (μ_m) is sufficiently high and the curvature (γ) is low, the positive term $\frac{m \cdot \gamma}{\mu_m}$ becomes smaller than the absolute value of the negative aerodynamic term (C_{lA}). This results in a negative denominator, forcing the solver to compute the square root of a negative number, which would yield a mathematically complex number and crash the real-domain simulation.

To prevent this, the algorithm implements a smart numerical safeguard. It calculates the theoretical friction threshold (mulim) at which the denominator becomes exactly zero:

$$\mu_{lim} = -\frac{m \cdot \gamma}{C_{lA}}$$

```
mulim = -Vehicle.Mass * gamma / Vehicle.ClA;
if mum > mulim
    mum = mulim;
end
```

Before computing V_{lim}, the solver checks if the actual mean friction (`mum`) exceeds this critical `mulim` threshold. If it does, the code artificially caps the friction coefficient to `mulim`.

3. **Speed Computation**
   The grip-limited cornering speed (V_{lim}) is derived from the steady-state equilibrium between the required centrifugal force for the current track curvature and the maximum lateral tire grip.

```
Vlim=sqrt(Vehicle.Mass*g/(Vehicle.Mass*gamma/mum+Vehicle.ClA));
Vehicle.Cl          = -1.2; % [] Lift coefficient
Vehicle.ClA=0.5*Air.Rho*Vehicle.Cl*Vehicle.A;
V= min([Vlim Vehicle.Vmax]);
```

Once the current limit speed of the vehicle is defined, the longitudinal acceleration required by the vehicle to switch from the previous step speed to the current one is computed.

$$a_{req} = \frac{V_{i+1} - V_i}{t}$$

The longitudinal acceleration is upper and lower-bounded, so the actual next step speed of the vehicle is limited by the capability of the vehicle of supplying the required acceleration.

```
% Vf = Vi + a*t
% Vi = x(6)
% Vf = xp(1,1)
% a = xp(6,1)
% t = 0.01 (time step)
xp(6,1) = max([-3*9.81 min([(V-x(6))/0.01 0.7*9.81])]);
xp(1,1) = x(6) + xp(6,1)*0.01;
```

The following flowchart resumes the logic behind the speed calculation for each step

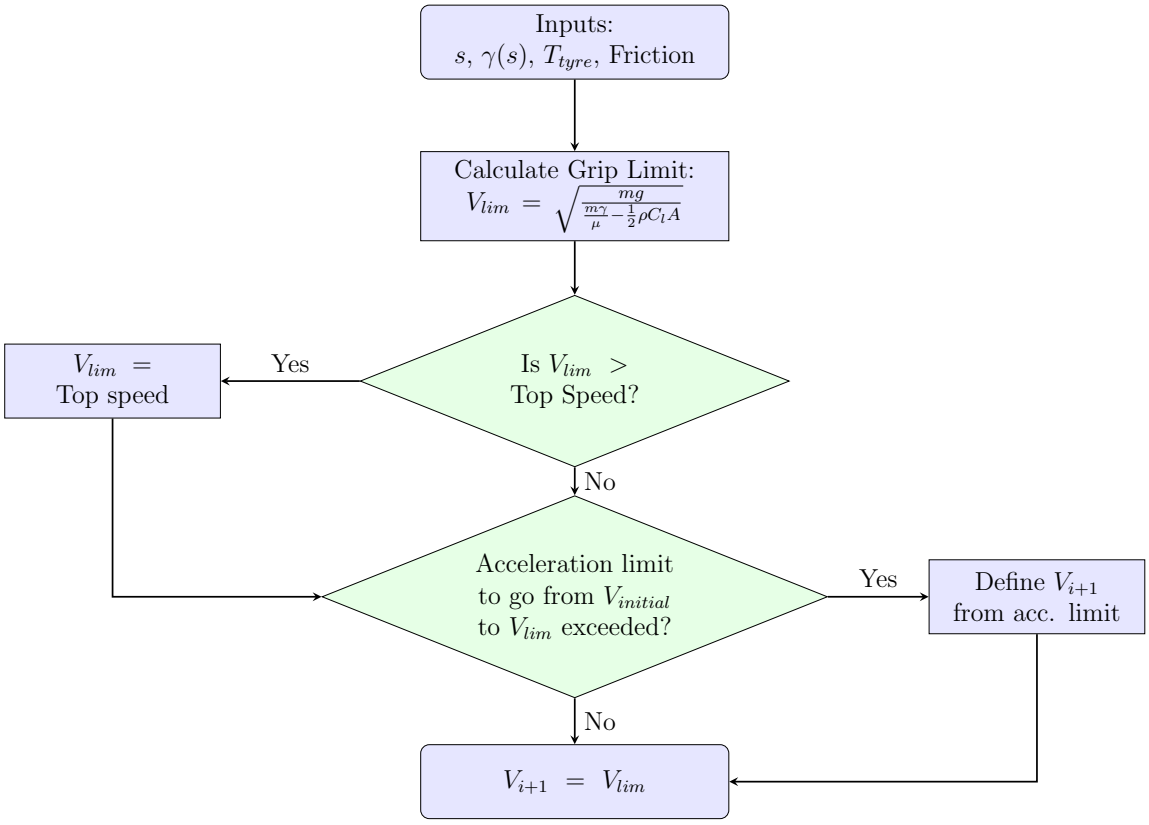


Figure 16: Velocity Limit Calculation

4. **Dynamics Section**
   First, the solver determines the required lateral acceleration based on the current track curvature:

$$a_y = V^2 \cdot \gamma$$

Vertical loads combine the vehicle's static weight and aerodynamic downforce, distributed via the Weight Distribution (WD) and Aerodynamic Balance (AB) parameters. Assuming longitudinal and lateral symmetry, the load on a single front tire is:

$$F_{z,front} = \frac{W_F - (F_{z,aero} \cdot AB)}{2}$$

Similarly, the total lateral force demand ($m \cdot a_y$) is distributed proportionally to the static mass. Thus, the lateral force required from a single front tire is:

$$F_{y,front} = \frac{m \cdot a_y \cdot WD}{2}$$

```
for c1=1:2 % 1 Front axle, 2 Rear axle

    % Lat. Forces + parasitic lat. force adjustment (toe angle)
    Fy(c1)=Fy(c1)+(Tyre(c1).Ca*Tyre(c1).CaFz*Fz(c1))*Vehicle.
   StaToe(c1);

    % slip angles
    alpha(c1)=Fy(c1)/(Tyre(c1).Ca*Tyre(c1).CaFz*Fz(c1))+Vehicle
   .StaToe(c1);

    Fx=0;
    if c1==2 % The car is Rear Wheel Drive

        % Long. Force + induced long. drag (tire scrubbing)
        Fx=(Vehicle.CdA*V^2)/2+Fy(1)*sin(alpha(1)+Vehicle.
   StaToe(1))+Fy(2)*sin(alpha(2)+Vehicle.StaToe(2));

        % Maximum Long. Force due to friction ellipse
        Fxmax=sqrt((Fz(c1)*mu(c1))^2-Fy(c1)^2);
        Fx=max([Fxmax Fx]);
    end

    % longitudinal slip
    csi=Fx/(Tyre(c1).Cx*Tyre(c1).CxFz*Fz(c1));
end
```

Within the iterative evaluation for each axle, the TMMT function calculates the specific forces and kinematic slip parameters (α and ξ) required to satisfy the vehicle's dynamic state. A crucial element in this formulation is the inclusion of the static toe angle (`Vehicle.StaToe`), which exerts a dual effect on the vehicle's dynamics:

- **Lateral Force Adjustment:** The toe angle introduces a parasitic lateral force even during straight-line motion. The model accounts for this by adjusting the required F_y.
- **Induced Longitudinal Drag (Scrubbing):** The tractive force must overcome not only the aerodynamic drag but also the "tire scrubbing" resistance generated by all four wheels. This induced drag is physically modeled as the longitudinal projection of the lateral forces ($F_y \cdot \sin(\alpha + \text{Toe})$).

Finally, the required traction is physically bounded by the limits of the friction ellipse (F_{xmax}), and the resulting longitudinal slip ratio (ξ) is computed.

5. **Thermodynamics Section**
   The final module of the `TMMT` function processes the kinematic slip and force outputs to evaluate the transient thermal balance of the tire.

   The temperature derivatives ($\dot{T}$) required by the ODE solver are computed by evaluating five distinct heat transfer mechanisms :

```
for c1=1:2 % 1 Front axle, 2 Rear axle
%% Thermodynamics
% Contact Patch Lenght cl
cl=Tyre(c1).CP.LC*Fz(c1)^0.7;
% Tire Lateral Surface Atot
Atot=pi*Tyre(c1).D*Tyre(c1).W;
% Aconv = Atot - A_ContactPAtch
Aconv=Atot-cl*Tyre(c1).CP.W;
% Heat Convection coefficient tread-air
hforc=Air.K*(0.0239(V*Tyre(c1).L/Air.nu)^0.805)/Tyre(c1).L;

cs=alpha(c1)/Tyre(c1).Alpharef*(Tyre(c1).Cs2-Tyre(c1).Cs1)+Tyre
    (c1).Cs1;
% Real Contact Patch Area due to corner deflection
Acp=cs*cl*Tyre(c1).CP.W;

% Frction Heat Generation Qfric
Qfric(c1)=Tyre(c1).Thermo.p1*V*(abs(Fx*csi)+abs(Fy(c1)*tan(
    alpha(c1))));
% Conduction to Track QcondTT (tread-track)
QcondTT(c1)=Tyre(c1).CP.HTT*Acp*(x(2*c1)-Track.T);
% Convection to Air QconvTA (tread-air)
QconvTA(c1)=hforc*Aconv*(x(2*c1)-Air.T);
% Internal Conduction QcondTC (tread-carcass)
QcondTC(c1)=Tyre(c1).Thermo.p3*Atot*(x(2*c1+1)-x(2*c1));
% Carcass deformation Qdef
Qdef(c1)=Tyre(c1).RRC*Fz(c1)*V;
end
```

   - **Frictional Heat Generation (Q_{fric}):** The mechanical work dissipated in the contact patch due to sliding friction acts as the primary heat source for the tread. It is a direct function of the longitudinal slip (ξ) and the lateral slip angle (α):

$$Q_{fric} \propto V \cdot (|F_x \cdot \xi| + |F_y \cdot \tan(\alpha)|)$$

   - **Deformation Heat Generation (Q_{def}):** The cyclic flexing of the tire structure generates internal heat within the carcass, modeled as a function of the rolling resistance:

$$Q_{def} = RRC \cdot F_z \cdot V$$

   - **Convection to Ambient Air (Q_{convTA}):** Heat dissipated from the tire surface to the surrounding air. The forced convective heat transfer coefficient (h_{forc}) is dynamically computed using a Nusselt number em-

pirical correlation, which scales non-linearly with the vehicle's velocity ($Nu \propto Re^{0.805}$).

- **Conduction to Track (Q_{condTT}):** Heat exchanged between the contact patch and the tarmac , scaling with the dynamic contact patch area (A_{cp}) and the thermal gradient ($T_{tr} - T_{track}$). The contact patch length is modeled to increase dynamically with the normal load ($F_z^{0.7}$).
- **Internal Conduction (Q_{condTC}):** The conductive heat flow transferring energy between the external tread and the internal carcass structure.

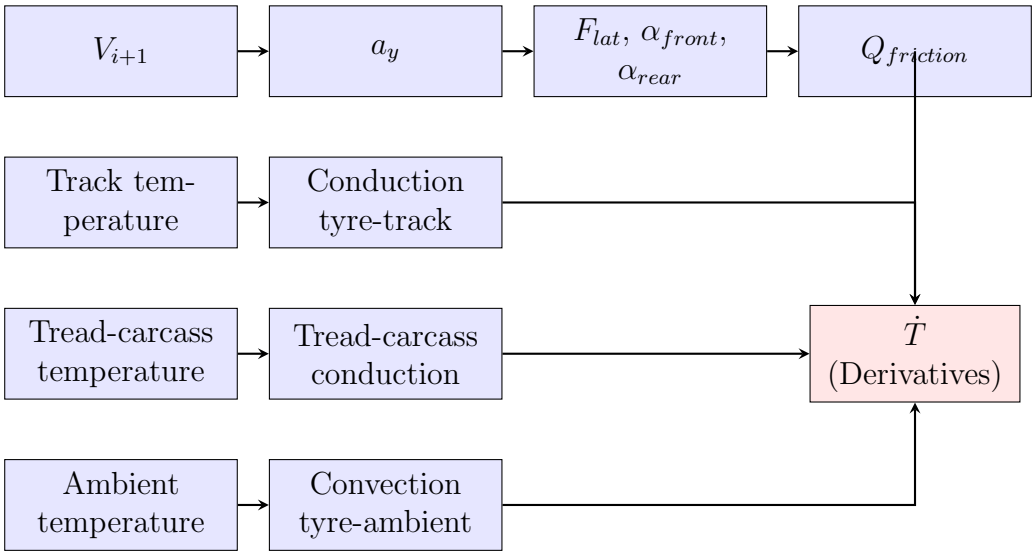


Figure 17: Heat Flows and Temperature Derivatives

2.2.2 Ambient simulation

The experimental setup consists of three different simulations evaluated across two track conditions (Cold Race and Hot Race), as summarized below. The environmental conditions for the simulations are defined by two distinct scenarios, tailored to assess the vehicle's thermal management and tire performance:

- **Cold Race:** Characterized by low ambient and track temperatures (e.g., Ambient: 15°C, Track: 25°C). These conditions typically delay the tire warm-up phase, making it challenging to reach and maintain the optimal thermal window. Consequently, the setup must favor heat generation to ensure adequate grip.
- **Hot Race:** Characterized by high ambient and track temperatures (e.g., Ambient: 30°C, Track: 50°C). In this scenario, the primary challenge is to avoid overheating, which could lead to a sudden drop in performance.

Simulation	Parameters Varied	Track Condition
Simulation 1	Front and Rear Toe	Cold Race
Simulation 2	Front and Rear Toe	Hot Race
Simulation 3	Weight Distribution & Aero Balance	Hot Race

Table 7

2.2.3 Toe Angle variation analysis in Cold Race

The first simulation was conducted to investigate the influence of toe setup on the vehicle's thermal management and overall dynamic performance. Specifically both the front and rear toe angles across three discrete values: $-0.5°$, $0°$, and $0.5°$.

The primary objectives of this simulation are :

- **Thermal Analysis:** To analyze the heat generation and distribution and thermal interaction between the tire carcass and the tread.
- **Performance Evaluation:** To quantify the impact of these thermal variations on Key Performance Indicators (KPIs), namely: best lap time, average lap time, and maximum speed (top speed).

To accurately isolate the individual axle contributions and identify any cross-coupling interactions, the post-processing of the results follows a structured, three-step methodology:

1. **Front Axle Sensitivity:** Analysis of the front toe variation while maintaining a fixed rear toe setup.
2. **Rear Axle Sensitivity:** Analysis of the rear toe variation while maintaining a fixed front toe setup.
3. **Combined Interaction:** Evaluation of the simultaneous variation of both front and rear toe angles to assess the combined effects on the overall vehicle balance and performance.

Front Axle Sensitivity:

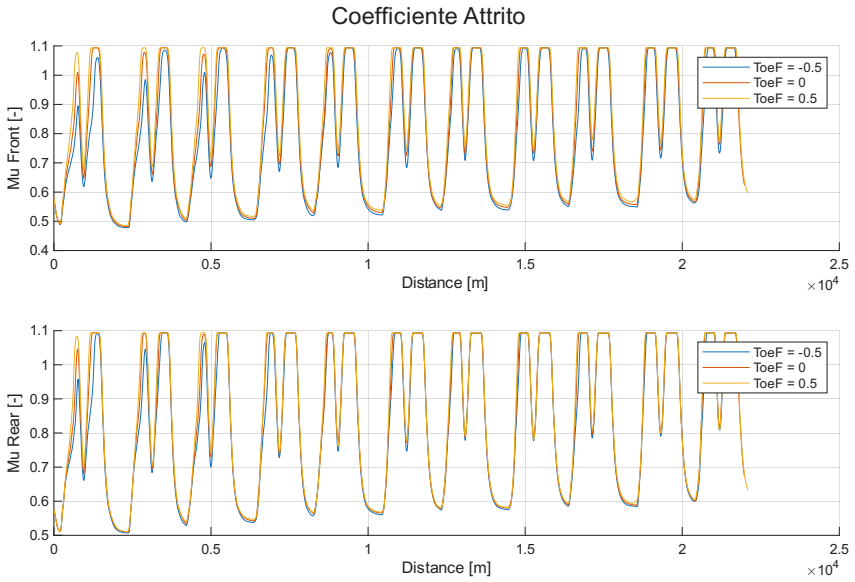
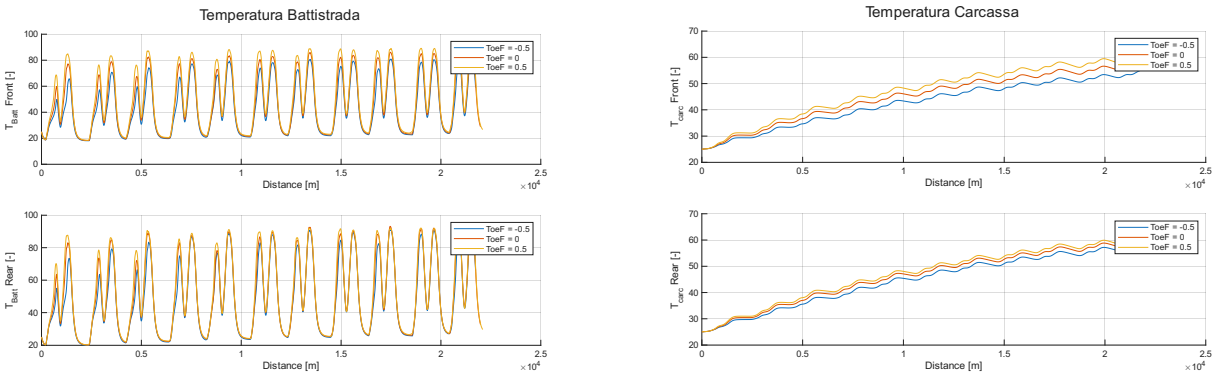


Figure 18



An analysis of the friction coefficient (μ) and temperature plots reveals substantial behavioral differences, particularly during the warm-up phase. The application of positive toe (toe-in) allows the tires to reach higher peak temperatures and rapidly activate, entering the operating window much earlier in the run.

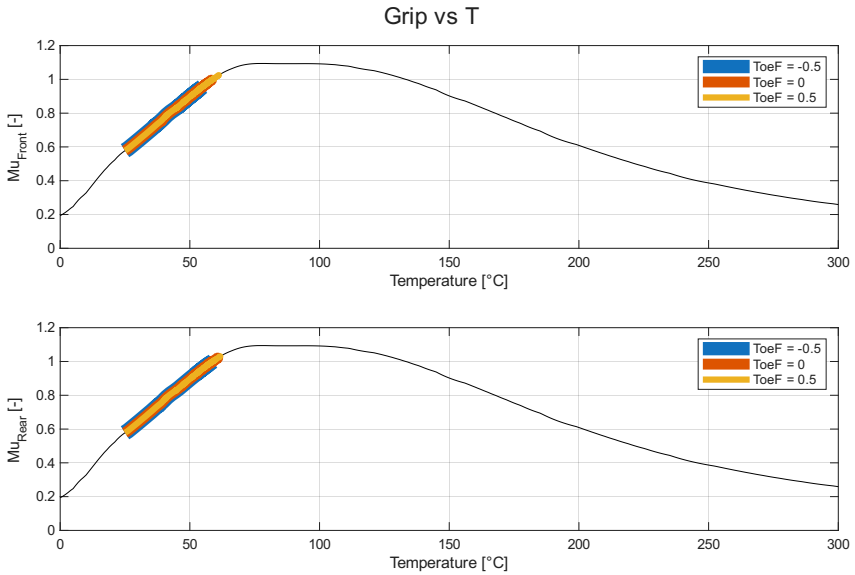


Figure 20

However, the thermal distribution profile (bell curve) clearly illustrates that all configurations fundamentally struggle with the low ambient and track temperatures. The data points are predominantly on the lower-temperature slope of the curve, indicating a persistent challenge in generating heat.

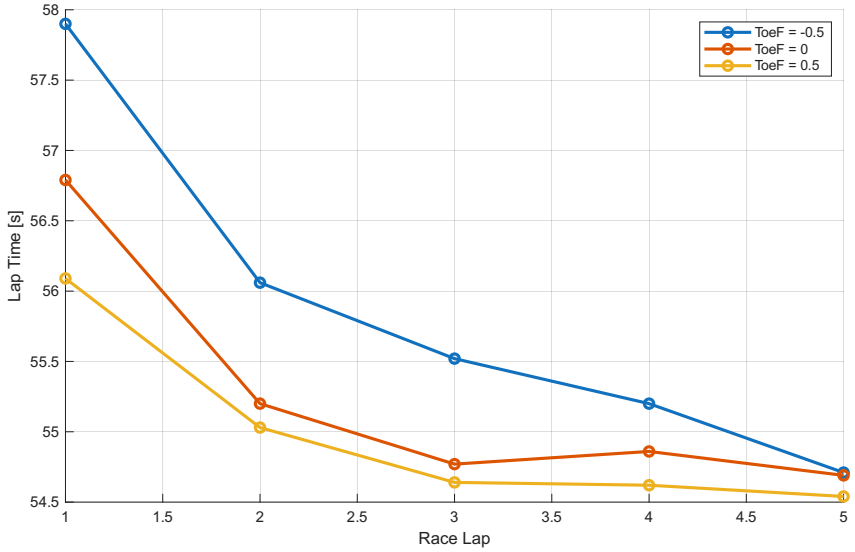


Figure 21

Consequently, the lap time progression exhibits a continuous decreasing (improving) trend throughout the simulation; as the stint progresses, the tires accumulate thermal energy, gradually increasing mechanical grip and lowering lap times. For the front axle, the positive toe configuration (0.5°) proves to be the most effective solution to counter this underheating.

Rear Axle Sensitivity An evaluation of the rear axle yields virtually identical conclusions. Both the thermal behavior and the lap time progression mirror the front axle's

trends, confirming that toe-in accelerates heat generation effectively across the entire vehicle.

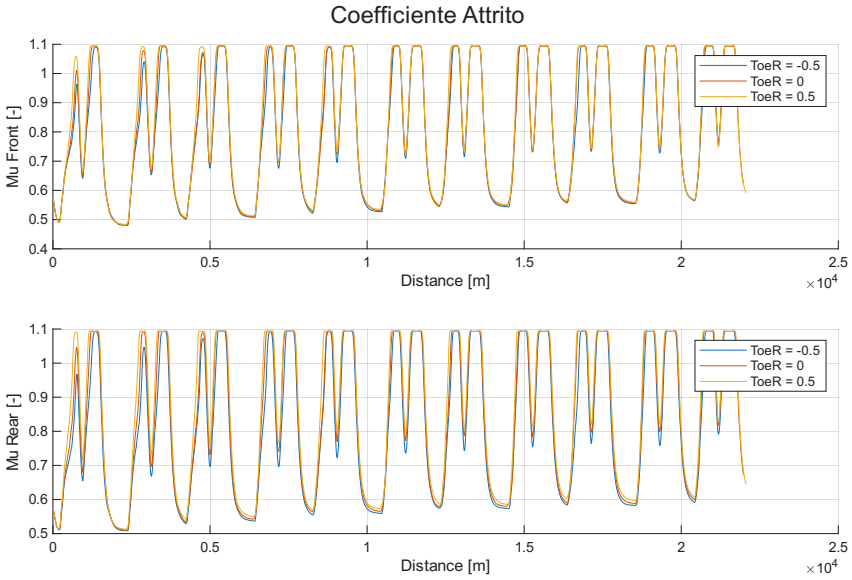
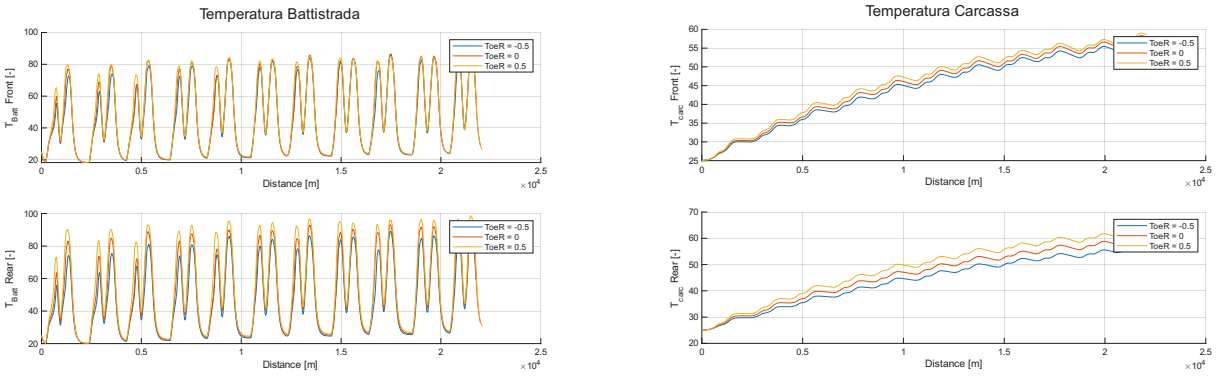


Figure 22



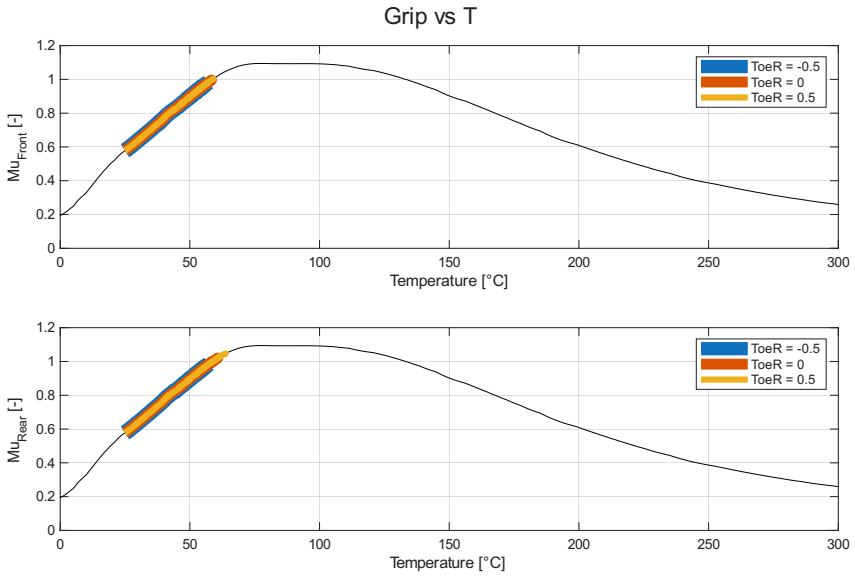


Figure 24

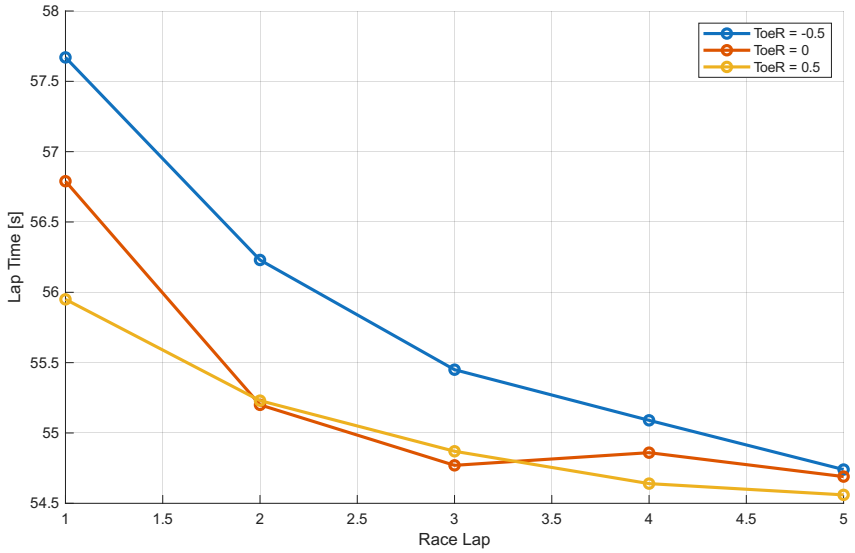


Figure 25

Combined interaction Moving to the combined axle analysis, the Maximum Speed matrix confirms the established aerodynamic baseline: the neutral 0°/0° setup yields the highest top speeds, while performance degrades at the extremities of the matrix due to elevated frictional drag.

However, unlike the *Hot Race* scenario, both the Best Lap Time and the Mean Lap Time are optimized when employing a combined positive setup (front toe-in and rear toe-in).

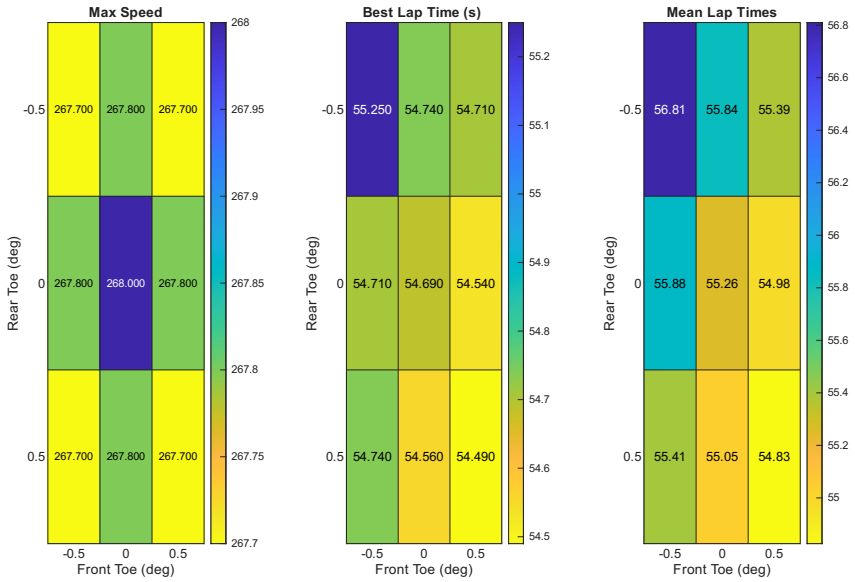


Figure 26

The rationale for this performance shift is explicitly detailed in the Combined Time in Window data (as shown in below). This metric demonstrates that the dual toe-in configuration achieves a significantly higher percentage of simultaneous thermal compliance compared to all other tested setups.

In these cold environmental conditions, the mechanical and aerodynamic drag penalties induced by the aggressive toe angles are completely offset by the massive mechanical grip advantage gained from rapidly heating both axles and sustaining them within their target thermal window.

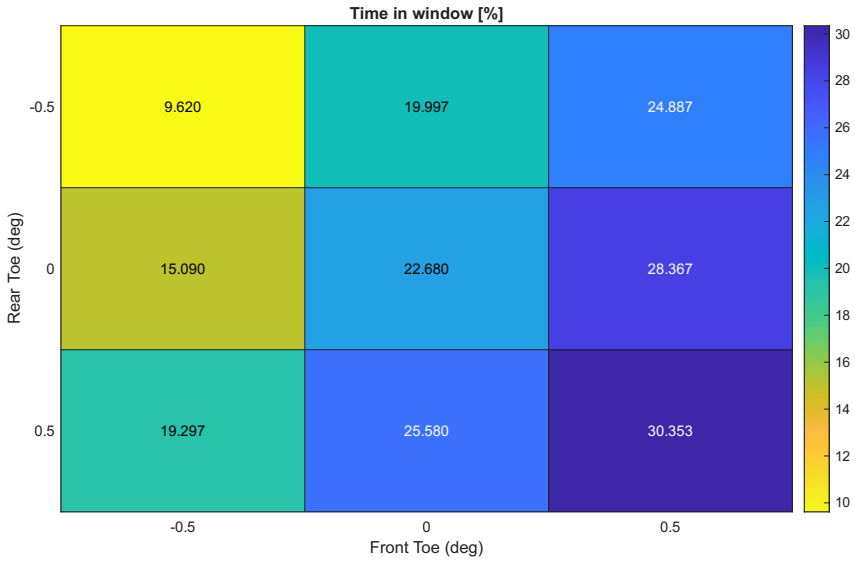


Figure 27

2.2.4 Toe Angle variation analysis in Hot Race

For the second simulation in hot race the same procedure was followed.

Front Axle Sensitivity

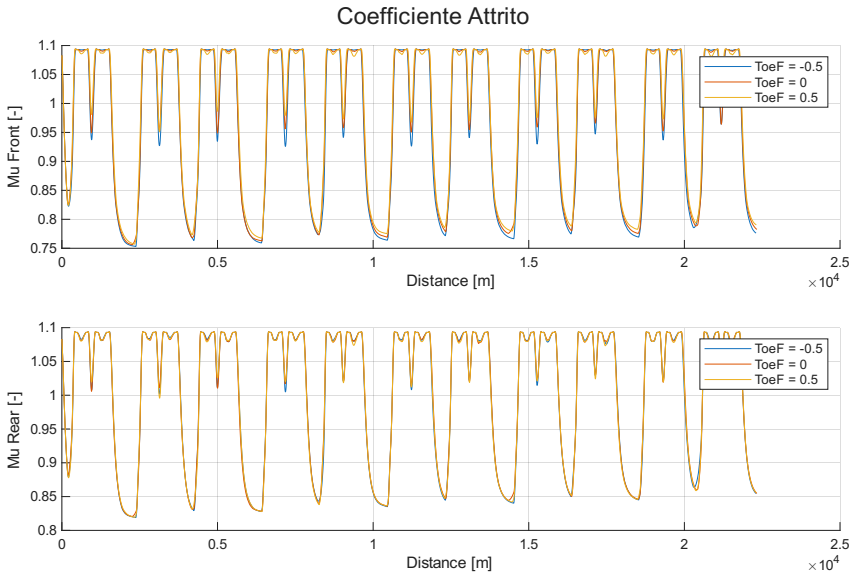
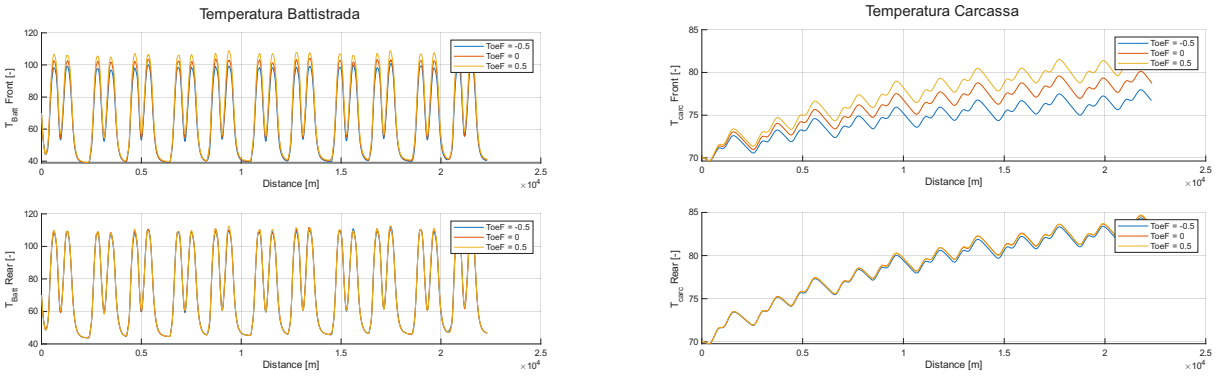


Figure 28



As evidenced by the Grip and Temperature plots, the application of positive front toe (toe-in) effectively mitigates excessive tire cooling along the straights, allowing the vehicle to reach the braking zones in a more favorable thermal state. However, under *Hot Race* conditions ($T_{\text{track}} = 50°\text{C}$), the overall tread temperatures are inherently elevated. The thermal distribution profile (bell curve) reveals that the positive toe setup forces the tires to operate consistently at the upper boundary of their optimal thermal window.

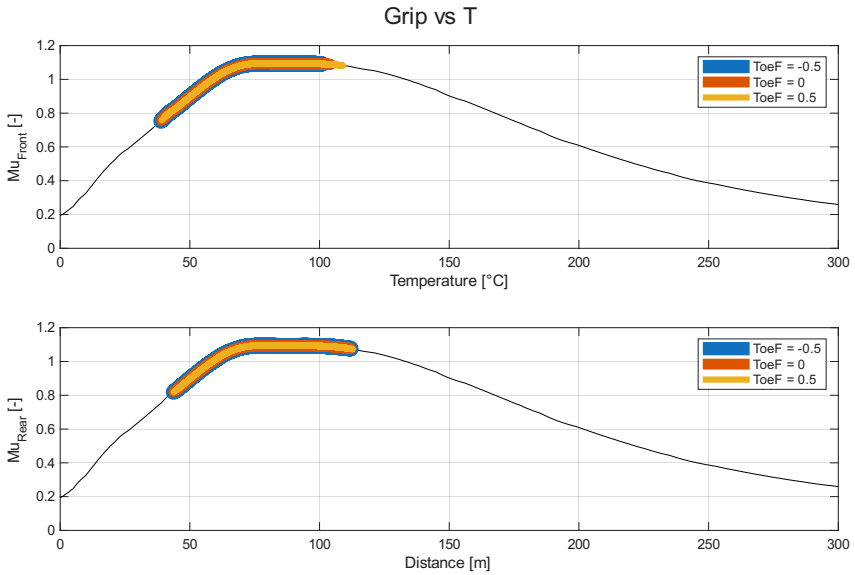


Figure 30

The thermal distribution profile (bell curve) reveals that the positive toe setup forces the tires to operate consistently at the upper boundary of their optimal thermal window.

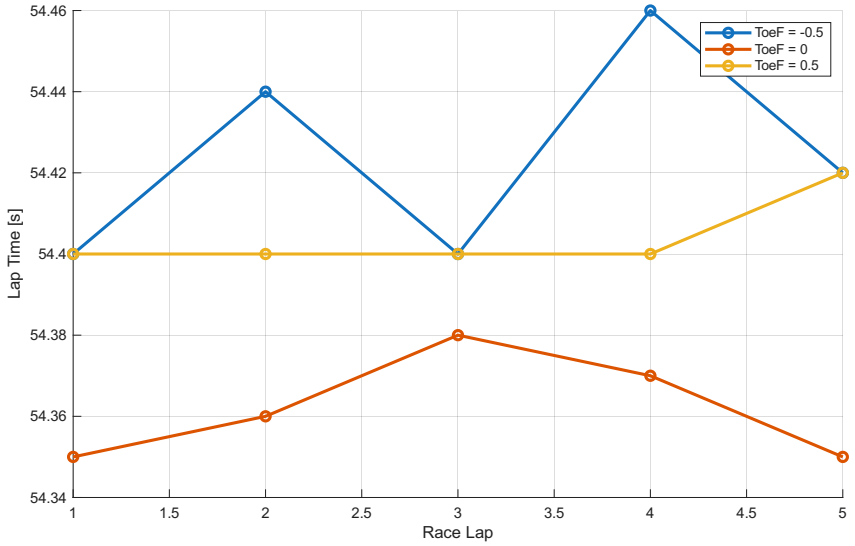


Figure 31

This thermal saturation is directly reflected in the lap time progression. Toward the end of the simulation, the lap times with a positive toe setup exhibit a degrading trend, primarily driven by tread overheating and a subsequent drop in mechanical grip. Conversely, the negative toe (toe-out) configuration demonstrates noticeable lap time oscillations, likely attributable to underheating and inconsistent thermal stabilization.

Ultimately, the simulation data suggests that a zero toe setup (0°) emerges as the optimal configuration for this specific environment. The inherently high ambient and track temperatures are sufficient to naturally sustain the tires within their operational window, while the neutral toe angle avoids inducing any frictional drag, thereby optimizing straight-line speed and preventing late-stint thermal degradation.

Rear Axle Sensitivity An evaluation of the rear axle data yields parallel conclusions to those observed during the front axle analysis. Under the elevated track temperatures of the *Hot Race* scenario, applying toe (whether positive or negative) to the rear wheels induces unnecessary frictional sliding.

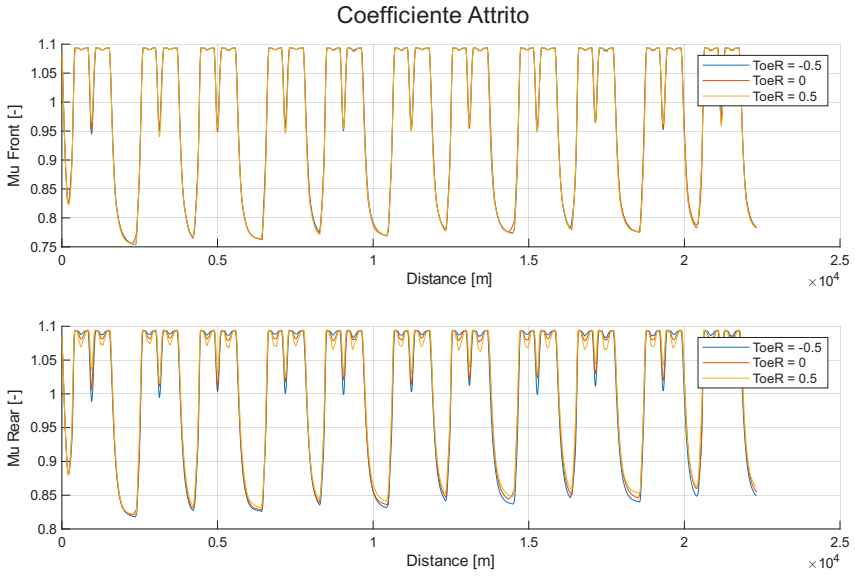
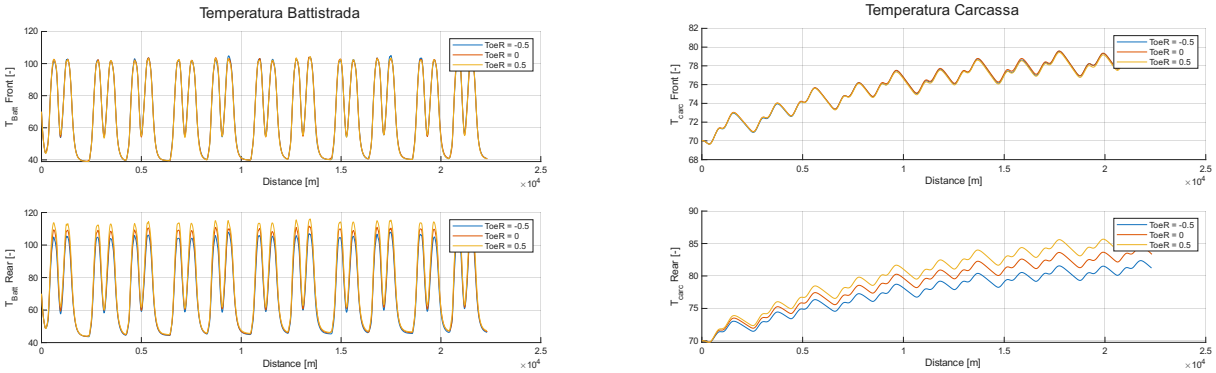


Figure 32



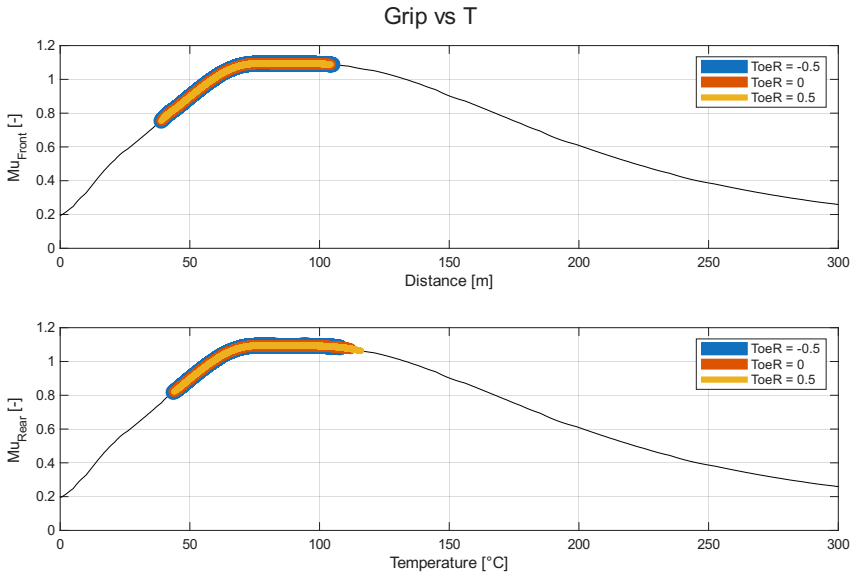


Figure 34

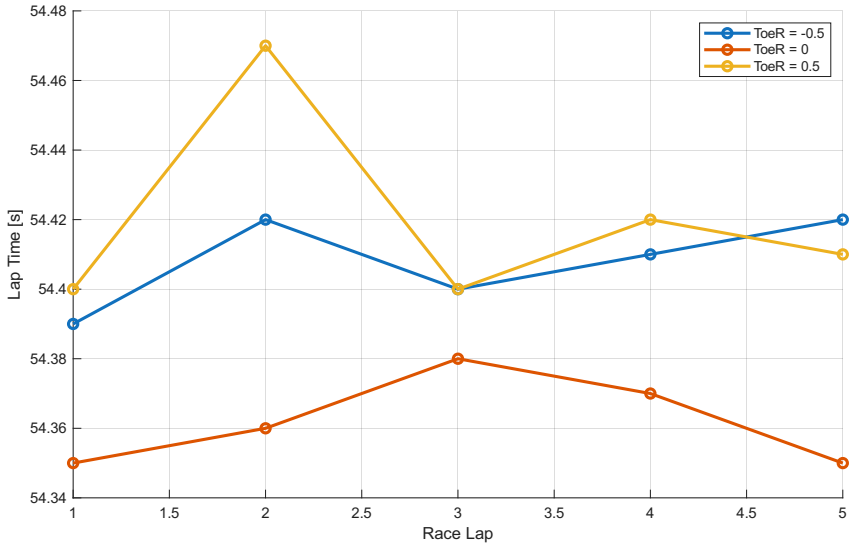


Figure 35

Consequently, the zero toe setup (0°) is definitively confirmed as the optimal configuration for both axles in this environment. By minimizing continuous slip, this neutral alignment ensures superior thermal stability across the stint, thereby maximizing mechanical grip and delivering the most consistent and competitive lap times.

Combined interaction To evaluate the combined effects of the setup changes under *Hot Race* conditions, the simulation results were aggregated into four primary metric matrices: Maximum Speed, Best Lap Time, Mean Lap Time, and Combined Time in Window. The latter is defined as the percentage of the lap during which both the front and rear axles operate simultaneously within the optimal thermal range of 70°C to 110°C.

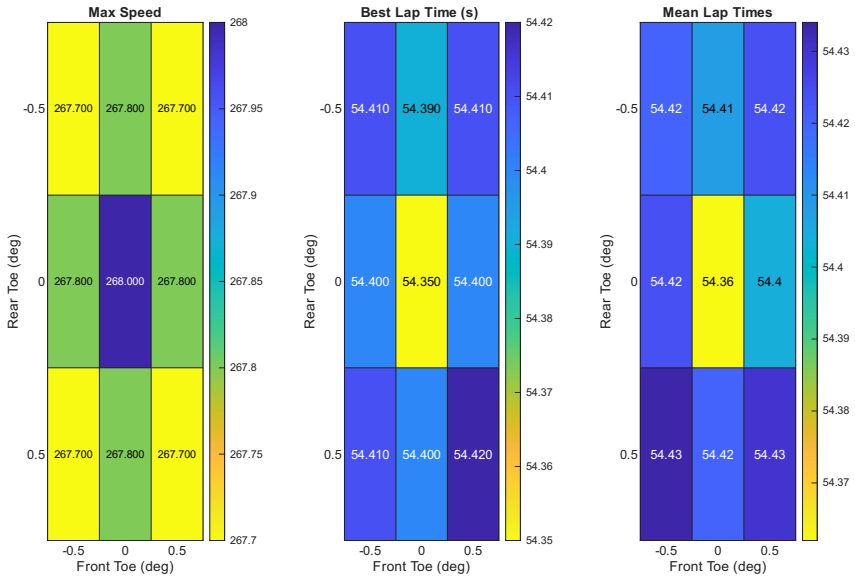


Figure 36

An analysis of the Maximum Speed matrix reveals that the neutral baseline setup (0° front and 0° rear toe) yields the highest top speeds. Conversely, the setups located at the extremities of the test matrix—representing the highest combined absolute toe angles—exhibit the poorest performance. This is directly attributable to the maximum frictional drag induced by the severe slip angles. This identical trend is mirrored in both the Best Lap Time and Mean Lap Time results. As established in the single-axle analysis, the elevated temperatures of the *Hot Race* scenario naturally sustain the tires. Consequently, the zero-toe configuration maintains the tires within the operational window long enough to completely negate any thermal generation benefits that toe adjustments might offer, while avoiding their associated drag penalties.

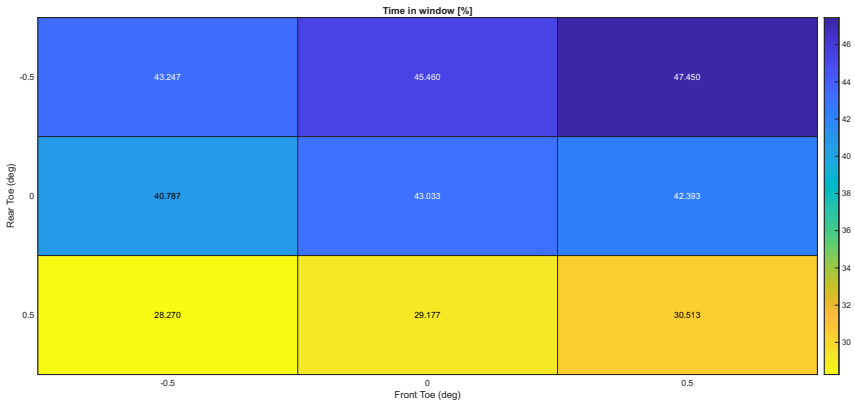


Figure 37

Interestingly, an examination of the Combined Time in Window data reveals a divergent optimum. The highest percentage of simultaneous thermal compliance is achieved with an asymmetric configuration: positive front toe (toe-in) combined with negative rear toe (toe-out). This specific setup effectively balances the vehicle's thermal state by forcing heat generation at the front (mitigating under-heating) while minimizing excess slip at the rear (preventing overheating).

However, despite maximizing the time spent in the optimal thermal window, this configuration does not translate to an overall performance advantage over the course of a 5-lap race. Furthermore the track is a high-speed oval, a layout where aerodynamic efficiency and high-speed dynamic stability are absolutely critical.

While asymmetric setups might theoretically optimize the *Combined Time in Window*, introducing rear toe-out (divergence) to control temperatures inherently destabilizes the rear axle. At the sustained high velocities typical of an oval circuit, this setup-induced yaw instability.

2.2.5 Weight Distribution and Aero Balance effect

This paragraph focuses on the variation of Weight Distribution ($WD\%$) and Aerodynamic Balance ($AB\%$) to analyze their direct impact on tire heating characteristics. Furthermore, the vehicle's performance is evaluated and compared in terms of top speed, fastest lap time, and average lap time.

To conduct this analysis, both the aero balance and weight distribution are varied using three distinct values each, creating a cross-combination matrix:

Minimum	**Medium**	**Maximum**
0.4	0.5	0.6

Table 8

The **Weight Distribution (WD)** represents the static division of the vehicle's total weight between the front and rear axles.

$$WD = \frac{W_f}{W_{total}} = \frac{l_r}{L}$$

Where W_f is the static weight on the front axle, W_{total} is the total weight of the vehicle, l_r is the longitudinal distance of the C.O.G from the rear axle, and L is the vehicle's wheelbase (the total distance between the front and rear axles).

In the context of the simulation, a WD value of 0.6 indicates that 60% of the total mass rests on the front axle, while the remaining 40% rests on the rear.

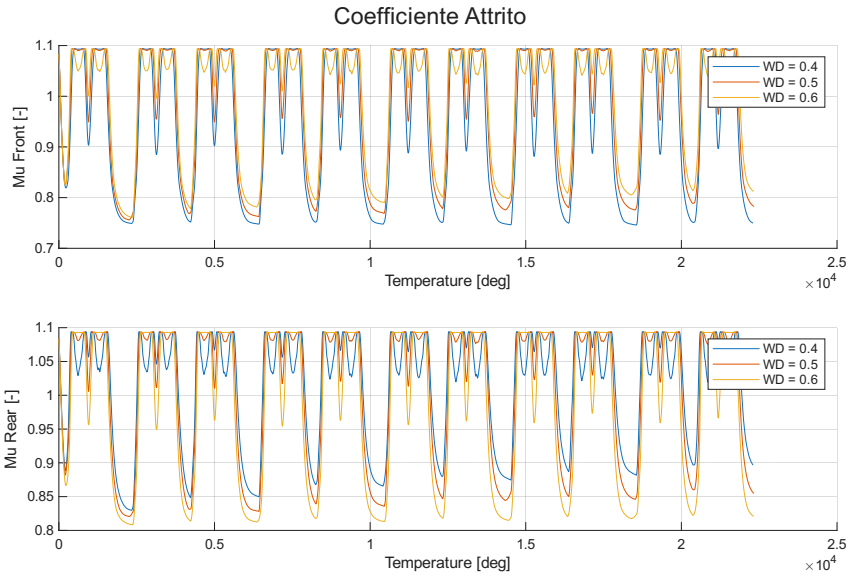
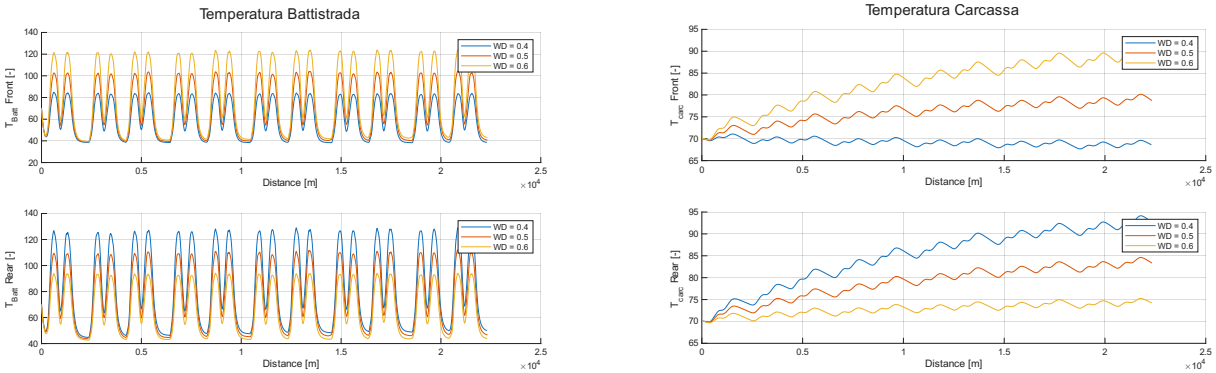


Figure 38



From the Tire Grip and Temperature evolution along the whole 5-lap race, it is clear how the Weight Distribution affects the under/over steering tendency of the vehicle.

A front loaded axle (WD = 0.6) slides more and so it is more subjected to over heating, especially in the high speed corner of the Indianapolis Race-Track, where the temperature increases so much that the front tires exits from their optimal temperature window, showing a decreasing trend of grip in the middle of the corner. The exact opposite same trend affects the real axle if the vehicle is rear loaded (WD = 0.4).

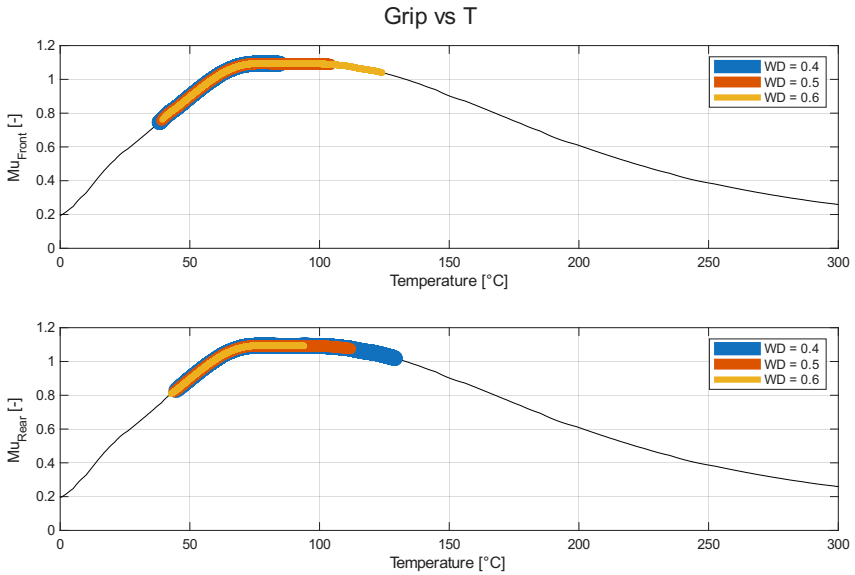


Figure 40

For what Concerns the LapTimes Performance on the given Race Track, the best setup in terms of Weight Distribution is the most neutral one (WD = 0.5). This is mainly because a central loaded vehicle doesn't induce sliding in a specific axle and so both the front and rear axles can operate in their optimal grip window, leading to a better overall RaceTime.

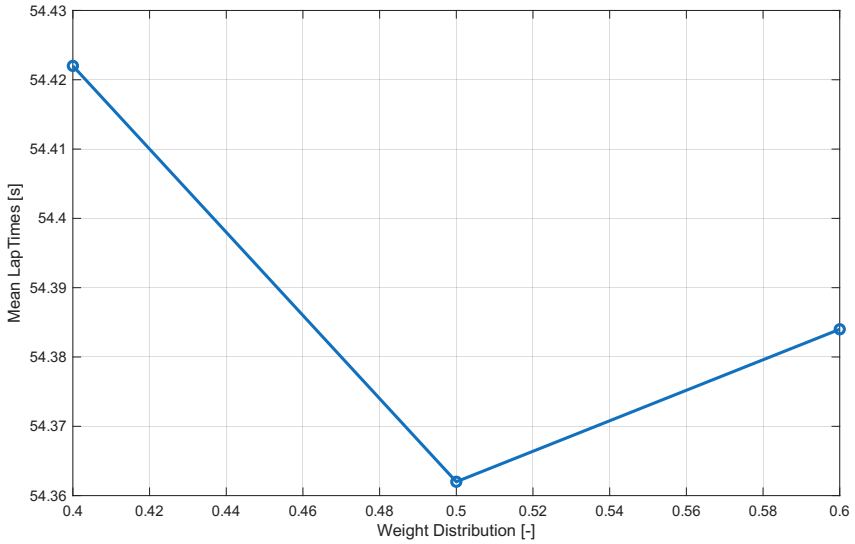


Figure 41: Mean LapTimes

The **Aero Balance (AB)** is defined as the percentage of the total aerodynamic downforce distributed to the front axle relative to the entire vehicle.

$$AB = \frac{F_{z,aero,f}}{F_{z,aero,total}} = \frac{C_{L,f}}{C_{L,total}}$$

Where $F_{z,aero,f}$ is the aerodynamic downforce on the front axle, $F_{z,aero,total}$ is the overall downforce, $C_{L,f}$ is the front lift/downforce coefficient, and $C_{L,total}$ is the total vehicle lift/downforce coefficient.

In the context of the simulation, a AB value of 0.6 indicates that 60% of the total downforce acts on the front axle, while the remaining 40% acts on the rear.

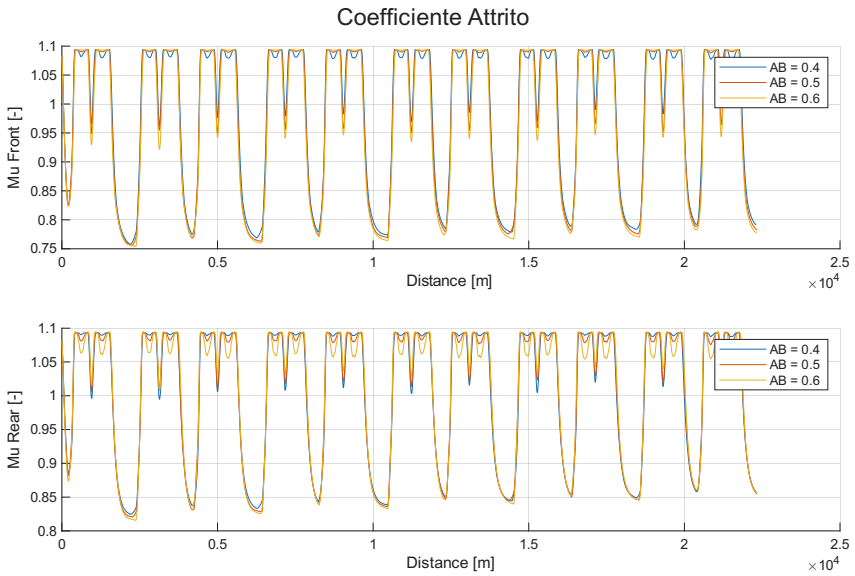
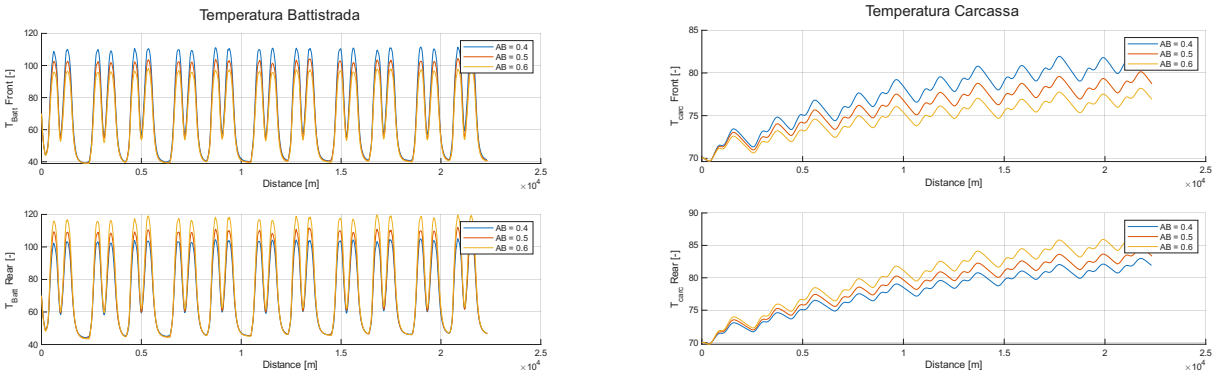


Figure 42



Unlike the Weight Distribution, a non-neutral Aero Balance provides more vertical load on one axle, preventing it from sliding. The frictional heating decreases and at the same time the hysteresis phenomena increases due to the higher amplitude stress cycle the tread is subjected to. As we know the hysteresis is the main actor in terms of grip generation, for this reason it's clear from the plots that an Aero Balance pointed towards an axle makes it more efficient in terms of grip generation and less exposed to over heating risks.

On the contrary, the "unloaded" axle will feel less aerodynamic load and so its sliding behavior will be more marked, over heating is more likely to happen.

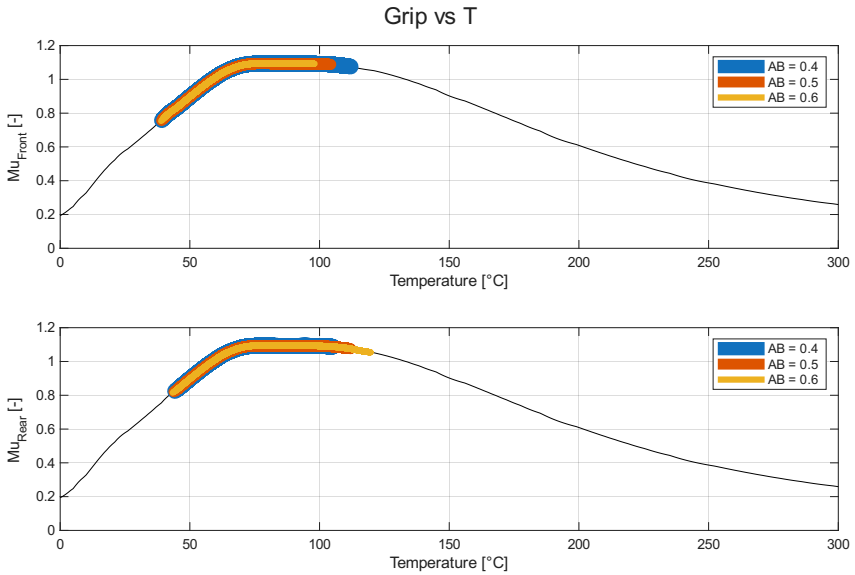


Figure 44

In terms of LapTimes Performance, the Aero Balance shows a similar trend to the Weight Distribution one. To maximize the total grip from both the axles, a balanced Aero Setup (AB = 0.5) is preferred.

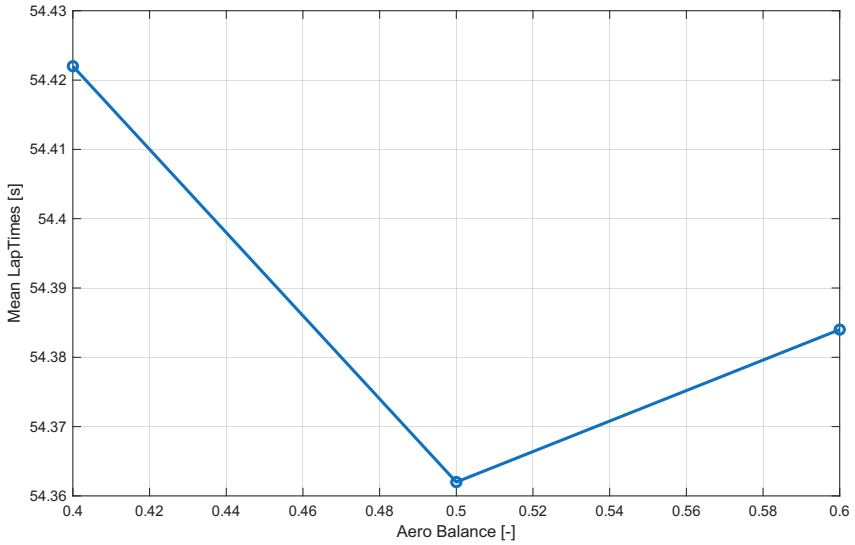


Figure 45: Mean LapTimes

Analyzing the **Combined Interaction**, we expect the two Setup Strategies to counteract each other. The **Best Setup** appears to be the **Neutral WD and AB Setup**, (WD = AB = 0.5). However similar performances can be achieved by counter-acting the effects of the two parameters: the more the car is unbalanced in terms of weight distribution, the more opposite aero balance is needed to load and neutralize the sliding axle

$$WD = 0.4 \rightarrow AB = 0.4$$

$$WD = 0.6 \rightarrow AB = 0.6$$

On the contrary the most unbalanced situations is exactly when the two parameters goes in opposite directions.

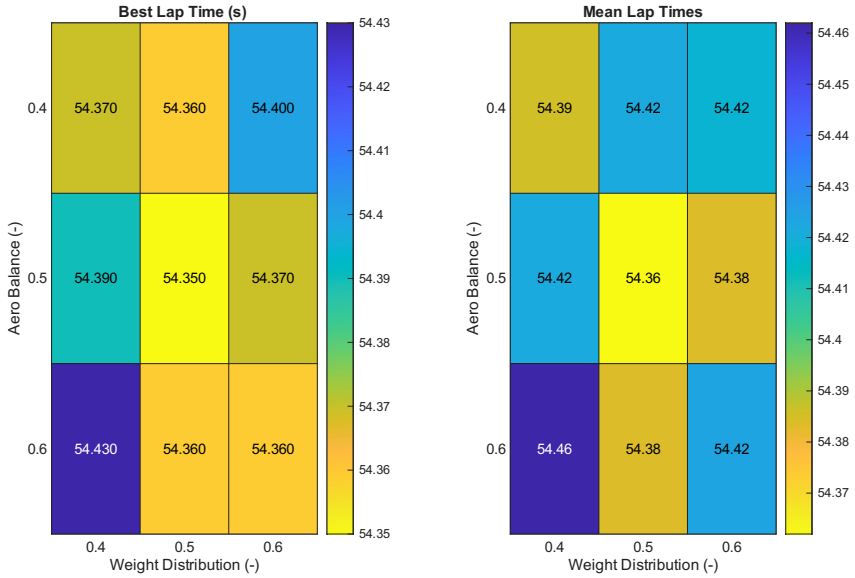


Figure 46

To quantify how much each axle generates heat from sliding, a custom performance metric has been introduced, designated as the **Sliding Index (SI)**.

During operation, the thermal energy generated by friction ($Q_{friction}$) acts directly on the tread. Due to its low thermal mass, the tread experiences rapid and violent temperature spikes. Conversely, the carcass absorbs this heat in a delayed and dampened manner via thermal conduction (Tread-carcass conduction). The instantaneous thermal gradient between the two layers is defined as:

$$\Delta T(t) = T_{tread}(t) - T_{carcass}(t)$$

To capture this thermal differences fluctuations over the entire race, the SI is defined as the statistical variance of this thermal gradient:

$$TSI = \mathrm{Var}(\Delta T) = \frac{1}{N-1}\sum_{i=1}^{N}(\Delta T_i - \mu_{\Delta T})^2$$

A low SI value indicates that the tread and carcass are operating in a stable thermal equilibrium, suggesting that the tire is rolling efficiently with minimal sliding. Conversely, a high SI value highlights that the tire is subjected to severe, localized cycles of transient surface heating.

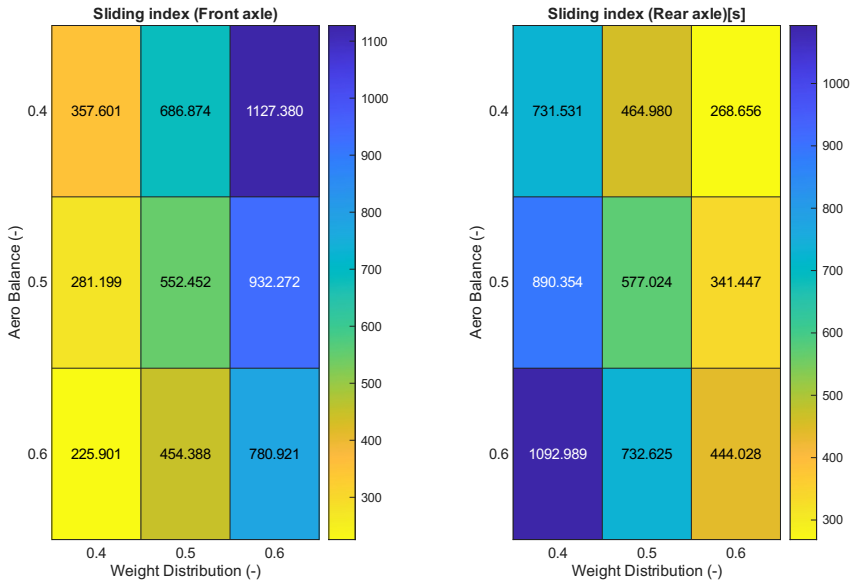


Figure 47

The tables shows a peculiar trend: to completely cancel the sliding effect on an axle, we would need to unload the tires from Weight Distribution and increase their Aerodynamic Downforce.

Axle	**WD**	**AB**	**SI Value**
Front	0.4	0.6	225.901
Rear	0.6	0.4	268.656

Table 9: Best SI index for Each Axle

However, reducing the sliding effect on one axle will increase drastically the sliding on the opposite one. Infact, to evaluate the overall performance of the vehicle, it is necessary to consider both the front and rear axles Sliding Indexes simultaneously. If we sum up the two contributions for each setup case, the **Neutral WD and AB Setup** confirms to be the optimal choice.

The final results of every different configurations of parameters are reported briefly in the table below.

Param.	Setup	Advantages	Disadvantages
Toe	+ (Toe-in) *Cold*	• Higher peak tire temps. • Faster warm-up.	• Creates drag/frictional force. • Reduces top speed.
	+ (Toe-in) *Hot*	• Prevents straightaway cooling.	• Overheats tires late stint.
	Neutral (0°) *Hot*	• No drag. Stable long-stint grip.	• Under-heats tires in cold race.
	Asymm. *Hot*	• Balanced front/rear heat.	• Rear yaw instability at high speed.
WD	0.4 (Rear-Hvy)	• Adds load to rear axle.	• Increases rear slip & overheating.
	0.5 (Neutral)	• Best lap time. Balanced grip.	• Bad if Aero Balance is offset.
	0.6 (Front-Hvy)	• Adds load to front axle.	• Increases front slip & overheating.
AB	0.4 (Rear-Aero)	• Prevents rear slip. Cools rear.	• Unloads front. Heavy front sliding.
	0.5 (Neutral)	• Max combined grip. Best baseline.	• Fails to fix mass (WD) imbalances.
	0.6 (Front-Aero)	• Prevents front slip. Cools front.	• Unloads rear. Heavy rear sliding.

Table 10: Compact summary of Toe, WD, and AB effects.

2.2.6 Alternative Tire Warm Up Strategies

A part from imposing a static toe angle on the wheels, there are many possibilities to warm up the tires in order to make them work in the optimal grip window.

- **Driving Techniques:**
  1. **Waving:** Performing slalom maneuvers can be useful in order put energy on the tire by generating prolonged load transfers. It is one of the most used techniques, however it can be decremental for tires wear.
  2. **Heavy braking:** The energy dissipated by the brake discs directly transferred through convection and radiation to the tire's inner carcass. The tire is warmed up from the inside, guaranteeing a more even heat distribution along all the layers.
  3. **Burnouts:** Exclusively done for the driven axle (normally the rear one), burnouts provide heat by causing violent wheel- spin.
  4. **Tire Warmers:** Tire Warmers are applied on the wheels when the vehicle is standing inside the box. The temperature can be set to keep the tires warm and ready to operate in their optimal grip conditions as soon as the vehicle leaves the garage.
- **Setups:**
  1. **Inflation Pressure:** Inflating pressure influences shape and extension of the contact patch: a low-pressure setting is preferred to extend the heat transfer area for sliding friction and increase the internal friction due to more deformations.
  2. **Camber Angles:** The camber angle primary affects the temperature distribution along the tire contact patch. A more uniform distribution (especially in corners) favors the tire total grip performance. A wrong camber setup my cause the tire to warm up not uniformly.

2.3 Exercise 2: Friction Law Identification

The aim of this exercise was to identify the tyre–road friction law from the contact data recorded during one lap. The available signals were the longitudinal and lateral contact forces, the vertical load, the sliding speed, and the tread temperature of the four tires. At each measured point, the friction coefficient was estimated as

$$\mu = \frac{\sqrt{F_x^2 + F_y^2}}{|F_z|}. \tag{21}$$

Three modeling strategies were considered. A polynomial response surface and a compact feed-forward artificial neural network were used as direct data-driven models of

$$\hat{\mu} = f(v_{\text{slip}}, T)\,. \tag{22}$$

A third approach used the Williams–Landel–Ferry (WLF) shift to interpret measurements at different speeds and temperatures as partial observations of a common reduced-speed master curve. The three methods were kept separate: the polynomial and ANN were used mainly for interpolation within the measured domain, whereas the WLF approach introduced a physical relation between temperature and excitation frequency.

2.3.1 Physical Filtering of the Telemetry

Raw telemetry includes steady cornering conditions, corner-entry and corner-exit transients, load transfers, and periods in which the tires are not close to their available friction limit. all fitting of all samples would therefore produce a model of average tire operation rather than a model representative of high friction utilization.

The first inspection also showed two distinct branches associated with the front and rear axles. This separation is mainly related to their different vertical-load and operating conditions. Since the selected regression models use only sliding speed and tread temperature as inputs, mixing both branches would neglect the load sensitivity of the friction coefficient. A homogeneous family was therefore obtained by selecting the loaded outer rear tyre during cornering. At every sample, the outer rear tyre was defined as the more heavily loaded tyre between RL and RR.

The physical filter required

$$0.95 \le \mu \le 1.40, \tag{23}$$

$$F_z \ge 5800\,\text{N}, \qquad \frac{F_{z,\text{outer}}}{F_{z,\text{inner}}} \ge 1.20, \tag{24}$$

and

$$\frac{|F_y|}{\sqrt{F_x^2 + F_y^2}} \ge 0.75. \tag{25}$$

The retained points were then grouped into continuous segments. Segments shorter than 15 samples were rejected, while five samples were removed from the beginning and end of each accepted segment. Finally, a local median consistency check was applied to μ, v_{slip}, T, and F_z in order to remove isolated outliers.

The filtering reduced the 501 outer-rear candidates to 243 points after the base physical criteria, 200 points after transient removal, and 193 points after the final consistency check. The selected dataset contained 127 RR samples and 66 RL samples. Its main ranges were

$$4.22 \leq v_{\text{slip}} \leq 8.65\,\text{m/s}, \tag{26}$$

$$64.6\,^{\circ}\text{C} \leq T \leq 164.3\,^{\circ}\text{C}, \qquad 6.39\,\text{kN} \leq F_z \leq 7.20\,\text{kN}, \tag{27}$$

with $0.969 \leq \mu \leq 1.261$.

Figure 48 shows the measured time histories and the samples retained by the filtering procedure. Figure 49 illustrates the front/rear separation, the residual load variation, and the coverage of the selected domain.

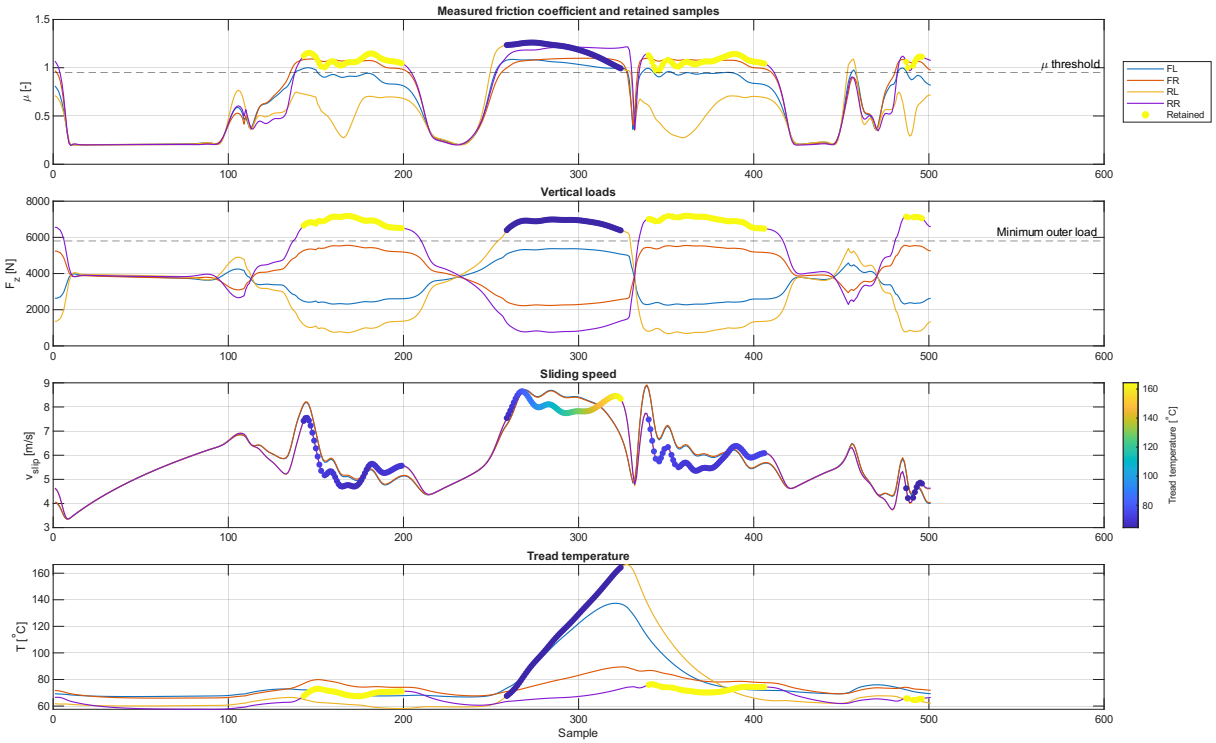


Figure 48Time histories of the measured friction coefficient, vertical loads, sliding speeds, and tread temperatures. The highlighted samples satisfy the physical filtering criteria.

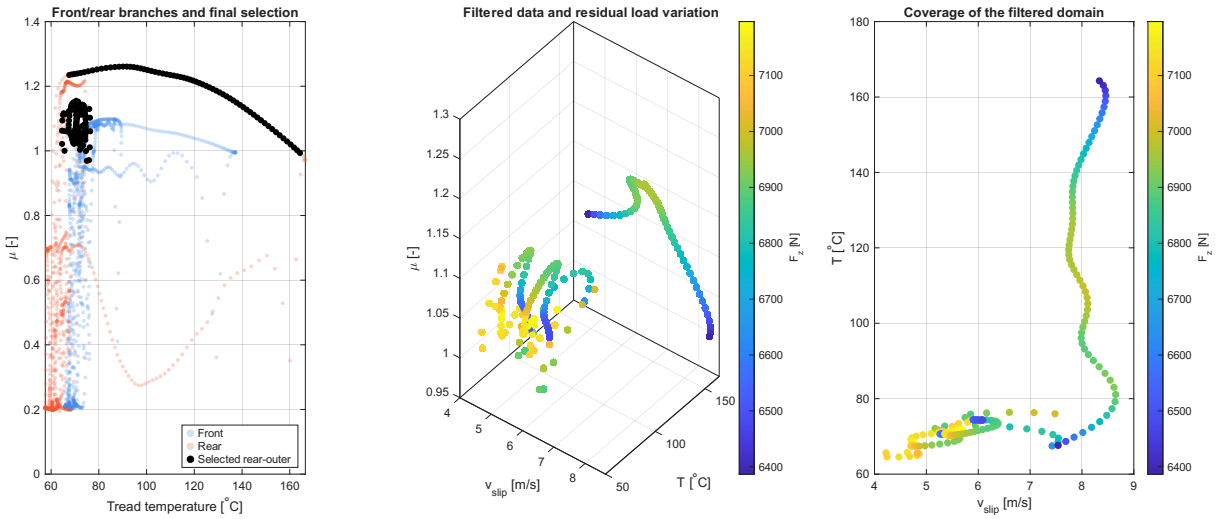


Figure 49Front and rear friction branches, selected outer-rear measurements, residual vertical-load variation, and coverage of the filtered domain.

2.3.2 Data-Driven Models

The filtered dataset was divided into a development subset and a held-out test subset. The test subset contained 20% of the retained measurements, corresponding to 39 points, and was stratified by tread-temperature range and selected rear tyre. The prediction error was calculated directly on the measured friction coefficient:

$$\mathrm{RMSE}_\mu = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\mu}_i - \mu_i)^2}. \tag{28}$$

The test error was not calculated on the wide V_{ref}–T_{ref} plotting grid. It was calculated only at held-out telemetry points. Consequently, it measures in-domain interpolation during the same lap and should not be interpreted as validation over unmeasured speeds and temperatures.

Polynomial response surface. The selected model was `poly32`, which contains nine coefficients and allows third-order dependence on sliding speed, second-order dependence on temperature, and the corresponding interaction terms. It achieved a cross-validated RMSE of 0.0293 ± 0.0038 on the development data. The model was selected because it provided the lowest mean cross-validation error and a clear improvement over lower-order surfaces.

Artificial neural network. The ANN was a feed-forward network with one hidden layer. Architectures with three to six hidden neurons and regularization coefficients from 0 to 10^{-2} were tested. Each configuration was trained five times with different initial weights. The lowest mean validation error was obtained with six neurons, but the three-neuron network was to the one-standard-error rule. The final configuration used three hidden neurons with no explicit regularization Its validation RMSE was 0.0257 ± 0.0041.

The model-selection results are summarized in Figure 50. The selected response surfaces are shown in Figures 51 and 52.

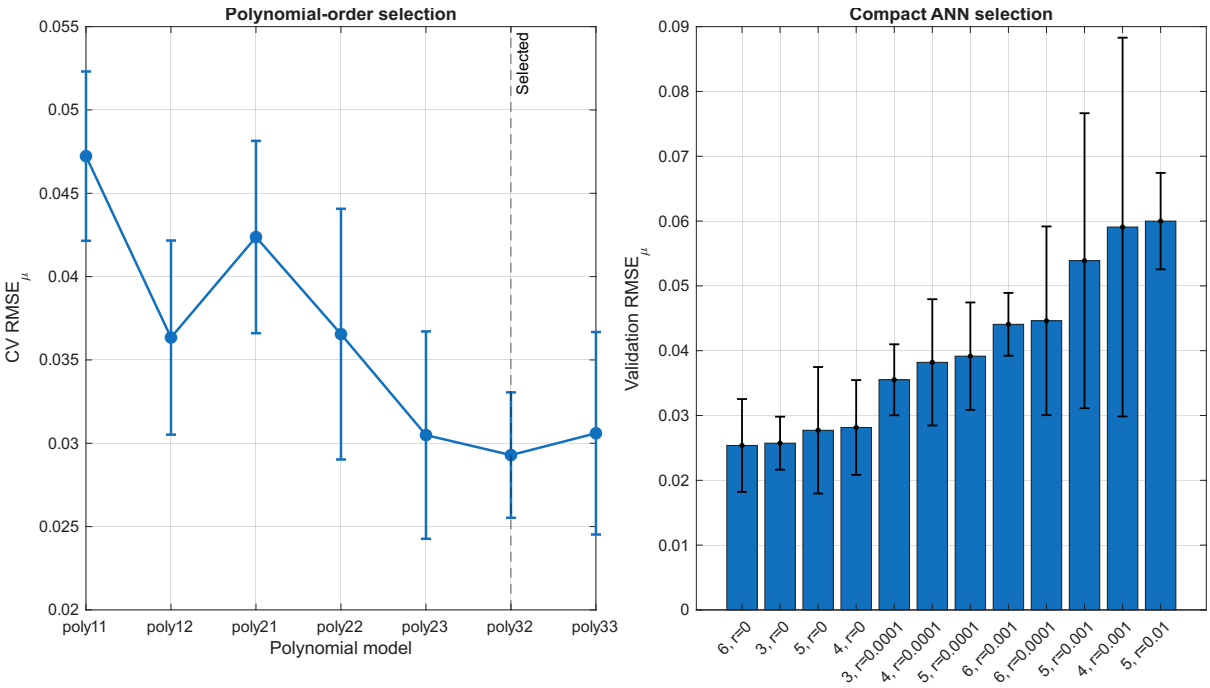


Figure 50Cross-validation results for the polynomial order, compact ANN architecture, and WLF spline smoothing parameter.

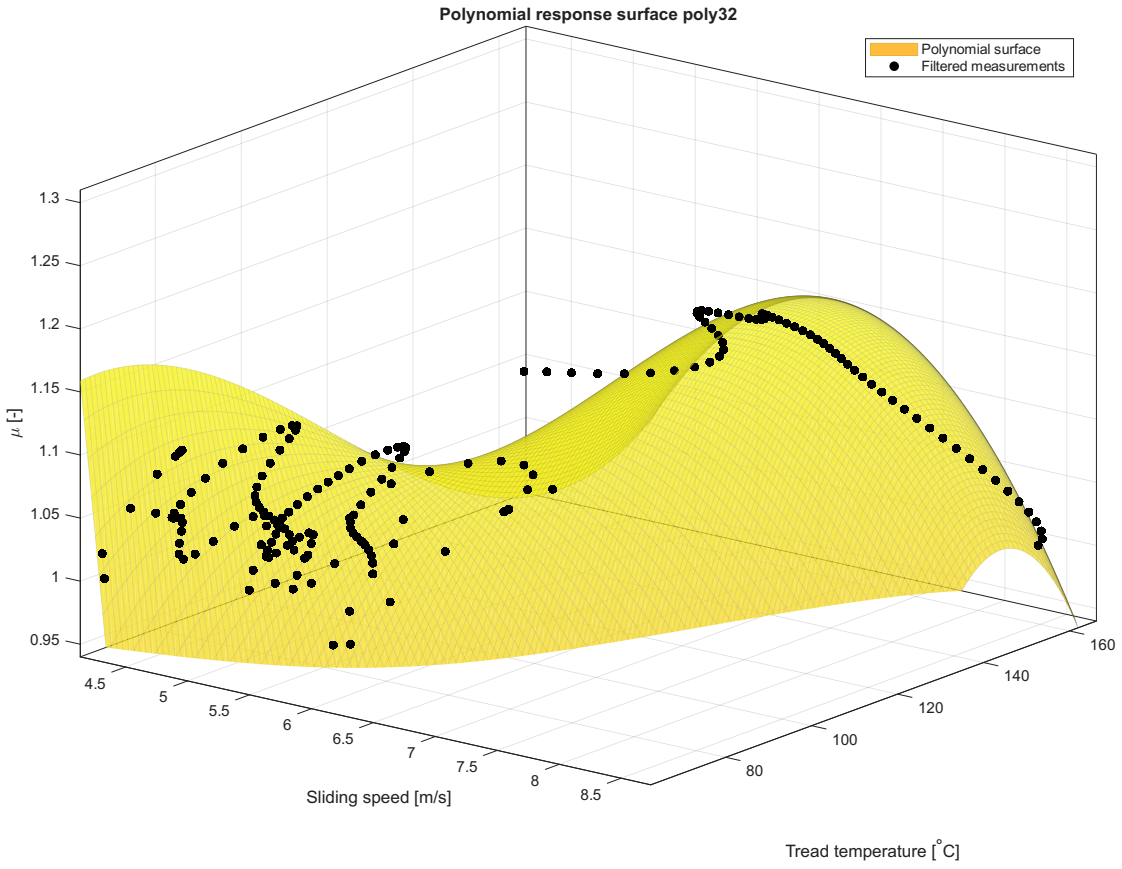


Figure 51Selected `poly32` response surface and filtered measurements.

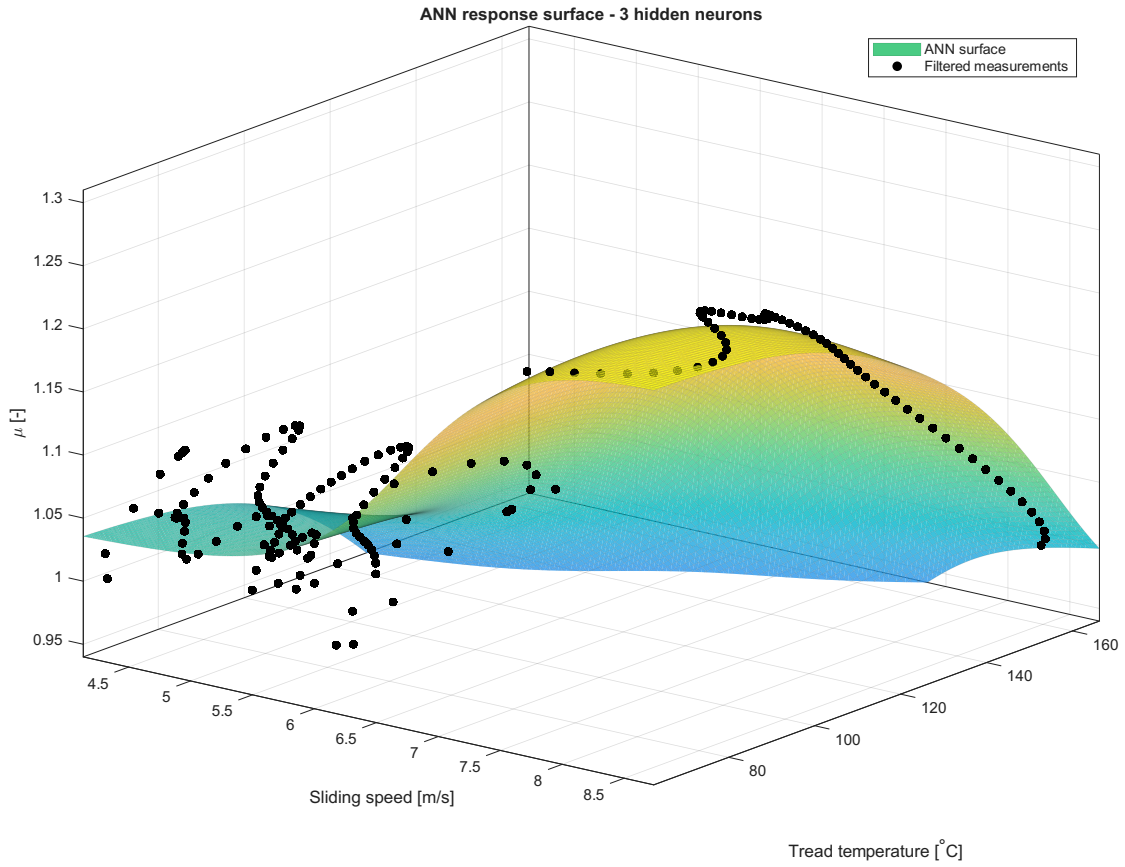


Figure 52Response surface generated by the selected ANN with three hidden neurons and comparison with the filtered measurements.

2.3.3 Fragment-Based Reconstruction of the WLF Master Curve

The data-driven models treat speed and temperature as independent regressor. The WLF method instead exploits the viscoelastic relation between temperature and excitation frequency. For a fixed characteristic road wavelength, the excitation frequency is proportional to sliding speed. Curves measured at different speeds may therefore represent different portions of the same underlying temperature-dependent response.

To test this interpretation, the filtered data were divided into narrow bands of sliding speed and vertical load. The adopted bin widths were

$$\Delta v_{\text{slip}} = 0.50\,\text{m/s}, \qquad \Delta F_z = 400\,\text{N}. \tag{29}$$

Each valid group had to include at least six samples and span at least 3 °C. A locally smoothed partial curve $\mu_k(T)$ was reconstructed for each group. Eight fragments were obtained. Their representative sliding speeds ranged from approximately 4.75 m/s to 8.32 m/s, while their median vertical loads ranged from 6.53 kN to 7.13 kN. Figure 53 shows these partial curves before shifting.

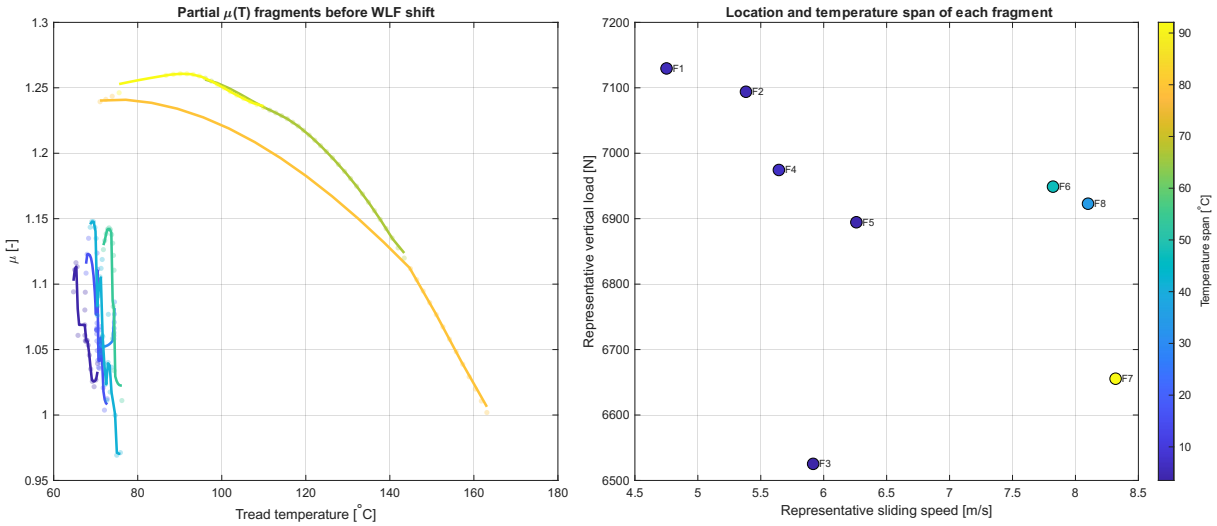


Figure 53Partial $\mu(T)$ curves reconstructed within narrow sliding-speed and vertical-load bands before the WLF shift.

The WLF shift factor was defined as

$$\log_{10} a_T = -\frac{C_1(T - T_r)}{C_2 + (T - T_r)}, \tag{30}$$

using $C_1 = 15$, $C_2 = 250\,^{\circ}\mathrm{C}$, and $T_r = 30\,^{\circ}\mathrm{C}$. Every point was mapped to the reduced coordinate

$$x_{\mathrm{red}} = \log_{10}(v_{\mathrm{slip}}) + \log_{10} a_T(T). \tag{31}$$

A smoothing spline was then fitted to the shifted fragments. The smoothing parameter was selected using leave-one-fragment-out cross-validation: a fragment is eliminated; same fragment is reconstructed translating the others; this process is repeated for the other segments. Although the lowest mean error occurred for a less smoothed curve, several candidates were statistically equivalent because of the large fold-to-fold variation. The smoothest eligible solution, $p = 0.50$, was selected to avoid following local irregularities. Its fragment-wise cross-validation RMSE was 0.0740 ± 0.0423.

Figure 54 shows that the fragments follow a common broad trend after shifting, but the collapse is not perfect. This is expected because RL and RR experience different thermal histories, the vertical load is only grouped within finite ranges, and the operating points may contain different levels of combined slip. Therefore, the WLF transformation captures the main temperature–speed trend, while part of the remaining scatter is due to variables not included in the reduced model. The final fragment reconstruction produced an RMSE of 0.0689 and $R^2 = 0.352$. The residual pattern indicates that speed and temperature alone do not explain the complete variation of the measured friction coefficient. Residual load differences, combined-slip conditions, road roughness over several wavelengths, and remaining transient effects are possible causes.

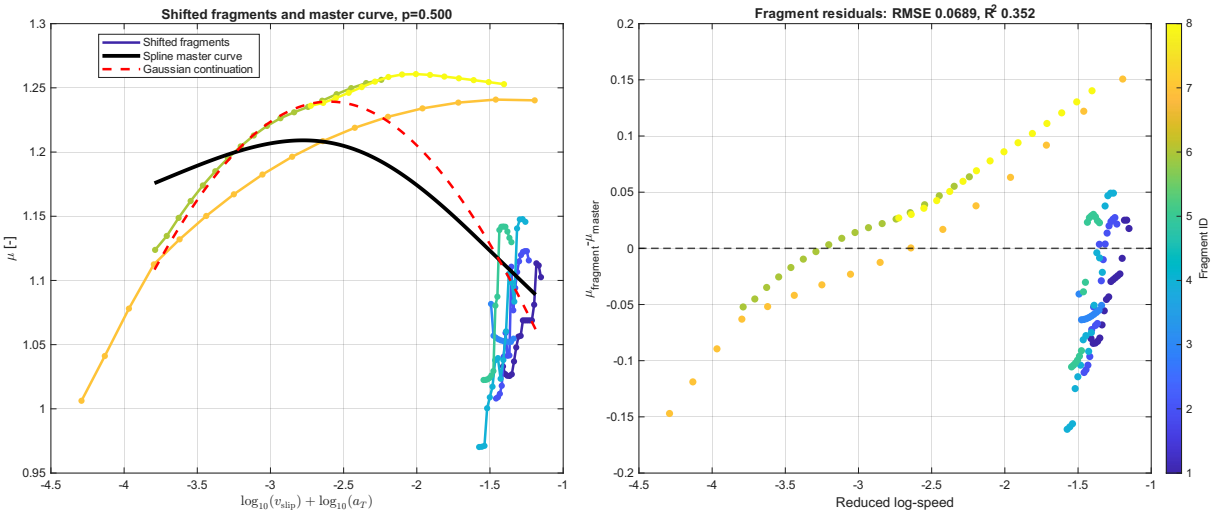


Figure 54Shifted fragments, smoothing-spline master curve, Gaussian continuation, and fragment residuals. The collapse captures the main trend but retains systematic dispersion.

2.3.4 Predictive Performance

The independent test results are reported in Table 11 and Figure 55. Three model are evaluated on the same 39 held-out telemetry points.

Table 11Prediction performance on the held-out telemetry subset.

Model	RMSE_μ	MAE_μ	R^2
Polynomial `poly32`	0.0280	0.0201	0.874
ANN, 3 hidden neurons	0.0273	0.0188	0.880
WLF smoothing spline	0.0611	0.0493	0.436

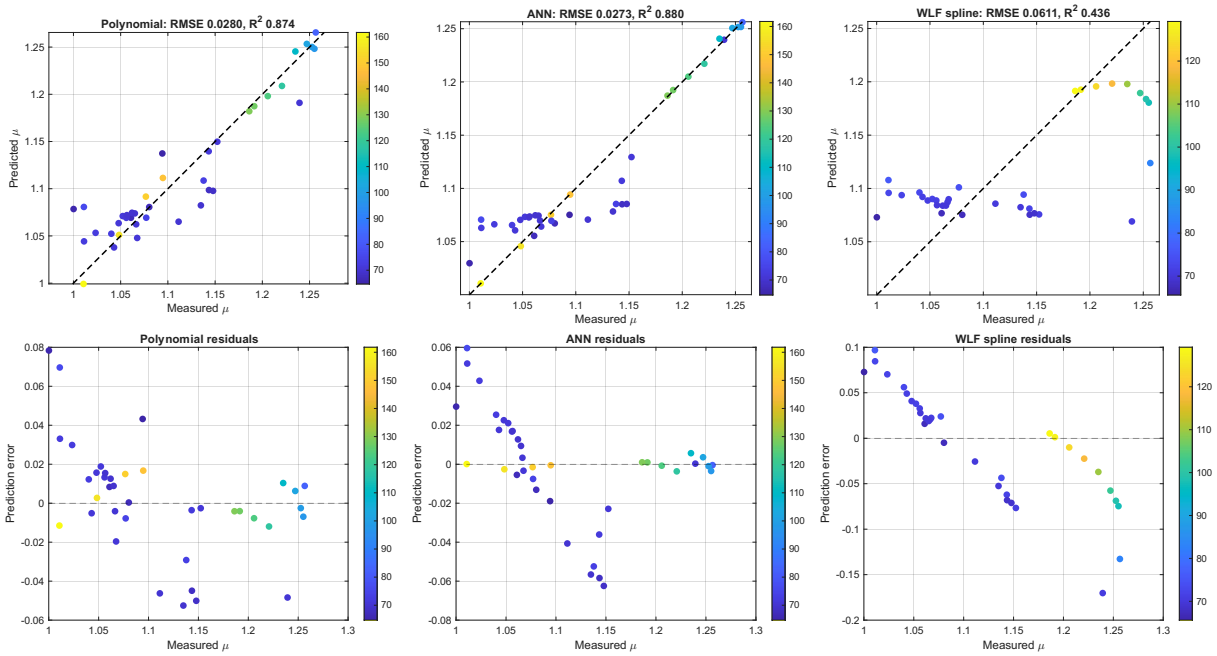


Figure 55Predicted and measured friction coefficients on the held-out telemetry subset, with residuals coloured by tread temperature.

The polynomial and ANN produced very similar errors. The ANN gave the lowest RMSE and MAE, but the difference from the polynomial was small. Both models reproduce the data well because the test samples lie inside the domain covered during the same lap. The residuals are also affected by the strong correlation between neighbouring telemetry samples. Therefore, the low errors demonstrate good interpolation, but not reliable extrapolation to the complete reference grid.

The WLF spline gave a larger error. This does not invalidate the reduced-speed interpretation, but it shows that a single WLF coordinate cannot fully represent all the friction variations present in the telemetry. The WLF model imposes a much stronger physical structure than the two flexible data-driven regressors, and this structure is only partially supported by the available fragments.

2.3.5 Data-Supported and Extended WLF Maps

The spline master curve was reconstructed over

$$-3.792 \leq x_{\mathrm{red}} \leq -1.189. \tag{32}$$

This interval can be mapped to speed–temperature pairs that were not directly measured, provided that their reduced coordinate remains inside the supported range. Such points use the WLF equivalence but do not require extrapolation of the master curve itself. The left panel of Figure 56 displays this data-supported map; blank regions require reduced-speed values outside the reconstructed interval.

The Gaussian was fitted to the binned points of the reconstructed spline master curve rather than to the raw telemetry. A robust nonlinear least-squares procedure with bisquare

weights was used to reduce the influence of the remaining local deviations between fragment, imposing the lower limit value as suggested a the beginning of the script provided

$$\mu(x_{\mathrm{red}}) = \mu_{\min} + A \exp\left[- \left(\frac{x_{\mathrm{red}} - b}{c} \right)^2 \right], \tag{33}$$

with

$$A = 1.1393, \qquad b = -2.5966, \qquad c = 3.4236, \qquad \mu_{\min} = 0.10. \tag{34}$$

The Gaussian is not imposed by WLF theory. It provides a smooth single-peak continuation outside the interval supported by the shifted fragments. The right panel of Figure 56 must therefore be interpreted as a model-based extension rather than a directly identified map.

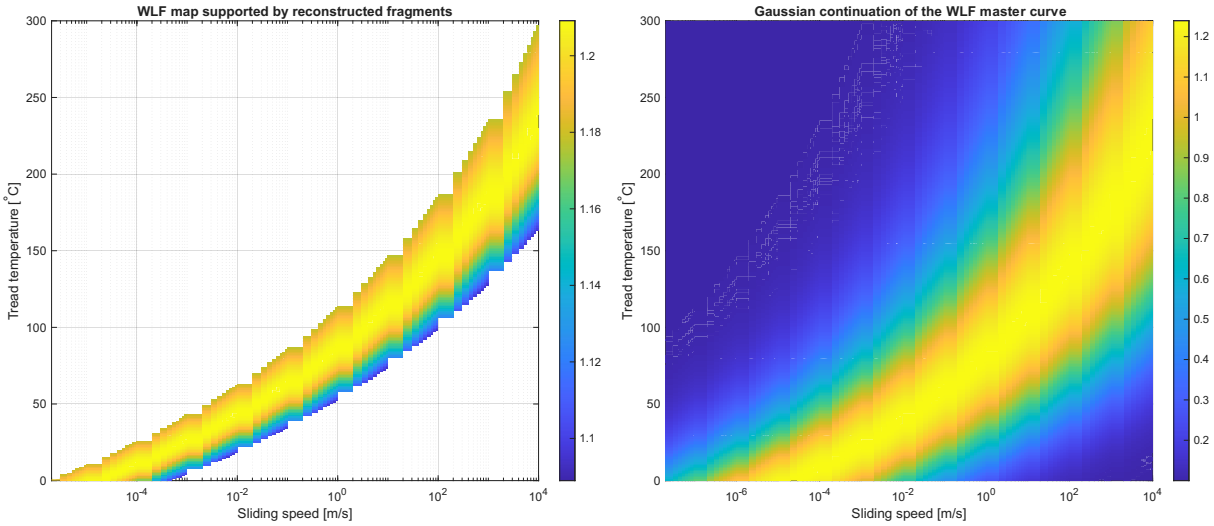


Figure 56Left: WLF map supported by the reconstructed reduced-speed interval. Right: Gaussian continuation outside the data-supported master-curve range.

Figure 57 presents temperature slices at fixed sliding speeds. Increasing sliding speed moves the high-friction temperature range towards higher temperature, since a higher temperature is required to compensate for the increase in excitation frequency. The solid spline portions are supported by the reconstructed fragments, whereas the dashed Gaussian portions include extrapolation.

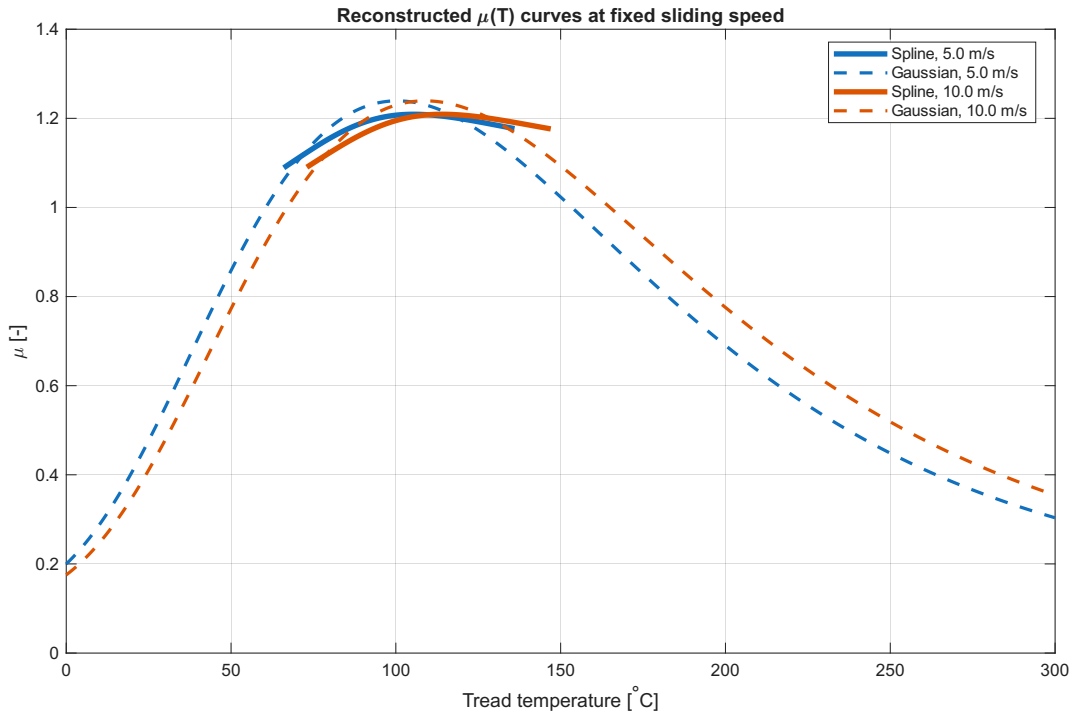


Figure 57Reconstructed friction curves versus tread temperature at fixed sliding speed. Solid lines denote the data-supported spline and dashed lines the Gaussian continuation.

2.3.6 Discussion and Conclusions

The three methods provide complementary information. The `poly32` surface gives a compact analytical representation and captures the main curvature of the filtered data. The three-neuron ANN reaches a slightly lower test error with a very small architecture. These models are the most accurate choices for interpolation within the measured operating domain.

The fragment-based WLF method addresses a different objective. It tests whether partial temperature curves obtained at different sliding speeds can be interpreted as sections of a common master curve. The shifted fragments show a shared trend, but the remaining dispersion and lower R^2 indicate that the available telemetry does not satisfy a perfect time–temperature superposition. This result is physically reasonable because the complete friction law also depends on load, road spectrum, combined slip, and other operating variables.

The WLF spline should therefore be considered the data-supported physics-informed reconstruction, while the Gaussian is only an optional continuation. The final comparison shows that physical filtering is essential before model identification, that polynomial and ANN models perform well for local interpolation, and that WLF shifting provides useful physical interpretation and structured extrapolation, although it is less accurate on the present test data.